

The Wheel of (Over)Time: Efficient Overtime Exchange via Coworker Networks

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Abstract

In the US public sector, there are many examples where a few government workers earn large amounts from overtime. Is this government inefficiency driven by insider influence, or an efficient reflection of worker preferences? We study the Los Angeles Department of Transportation, where several traffic officers earned around \$100,000 in overtime pay despite an ex ante equal system called “the wheel.” Consistent with the existence of insider influence, well-connected officers appear to exploit their informal network on a secondary internal market for shifts. Nevertheless, informal trading achieves 93.6% of the maximum possible allocative efficiency. This occurs because preferences across officers are an order of magnitude larger than trading frictions. Even if trading frictions were much larger, the informal trade network encodes the preferences of officers because it is formed through work. This illustrates a general idea: endogenous trading networks can create the conditions for the Coase Theorem to hold.

Keywords: informal trade, auctions, overtime, networks

JEL codes: J22, M5

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1 Introduction

Public sector organizations in the United States are often accused of excessive and unequally distributed overtime, with news articles reporting examples of government workers making tens and sometimes hundreds of thousands in overtime pay alone (Sedacca 2024, Tchekmedyian and Pringle 2025, Nelson 2015). At the same time, overtime is often allocated informally within these organizations. Is the concentration of overtime among a few individuals a sign that public resources are captured by influential insiders, or an efficient consequence of preference differences across workers? The answer has implications for the design of overtime systems, in particular whether reductions in overtime costs and inequality are possible without severely reducing worker surplus. Given the sheer magnitude of observed differences in overtime in publicized examples, and that markets rarely operate within firms, much less government organizations, we might expect influence to rule in efficiency’s stead.

This paper demonstrates that an informal, influence-based institution inside a government organization can make relatively efficient allocations, and that the concentration of overtime in the hands of a few can reflect preference differences across workers. We study the Los Angeles Department of Transportation (LADOT), which assigns overtime equally initially using a rotating list system called “the wheel,” but allows officers to trade informally. Despite initial equality of opportunity, a small number of traffic officers earned more than \$100,000 in overtime pay in a 1.5-year period. A government report found (Galperin 2015) and the *Los Angeles Times* reported (Nelson 2015) that this was the result of a “select few employees with insider relationships.” We obtained and made public record fine-grained pay records which allow us to identify when officers worked in the same location on the same day and therefore recover the network of potential relationships within the organization. We show that an officer is several times more likely to work overtime when they are well-connected to the coworkers who were likely initially endowed with overtime, confirming that influence within the organization provides better access to overtime.

However, the inequality generated under informal trade turns out to be efficient, in that it largely reflects differences in preferences for overtime across workers rather than differences in insider access. The LADOT pairs essentially fair ex-ante assignment of overtime with an informal secondary market for overtime where shifts can be traded or sold without management intervention. These features, combined with the daily panel structure of the data, allow us to separate access from preferences. Decompositions reveal that removing differences in informal networks has a negligible impact on overtime inequality, while removing preference differences significantly reduces inequality. We further confirm this by counterfactually replacing informal trading with formal, centralized auctions where officers bid wages for shifts.

Although uniform-wage auctions achieve the maximum possible allocative efficiency, informal trading achieves 93.6% and has essentially equal allocative efficiency as a uniform-markdown auction where officers bid markdowns from their union-negotiated wage. The counterfactual auctions also have no impact on or worsen inequality. The overtime wage savings from moving to these counterfactual auctions are modest, but the allocative efficiency lost from moving from informal trading to random assignment is large and equivalent to \$4.32 million in lost earnings.

Our results rest on our ability to separate preferences from access to overtime. In many settings this is challenging; either the systems used to assign work are unclear and complex, the employment contract and tasks vary across workers within the organization, or overtime assignments are dictated by managers and thus do not contain information about worker preferences. By studying Los Angeles traffic officers, we resolve all three challenges. The overtime assignment process, colloquially referred to as the wheel, is well understood by officers and can be accounted for using our data. There are over 400 active traffic officers at any given time, all of whom share the same occupation, job title, tasks, and pay scale. These details of the job are well-documented in union-negotiated contracts. Conditional on volunteering, all officers are initially entitled to the same amount of overtime. Finally, and perhaps most importantly, they are allowed to buy, sell, or trade their overtime via an informal market that is unregulated by management. Officers desire overtime both because it is paid at 1.5 times their normal hourly wage, and because most overtime involves special events, which frequently feature professional athletes and celebrities.

The key technical challenge in our setting is accounting for the informal market. An officer who is observed working a large share of overtime may be doing so either because they enjoy working overtime or because they are well connected within the organization and have better access to overtime. We account for the informal market by tracking all the instances where two officers worked on the same day in the same location. For 90-day rolling windows prior to each work date, we sum up this measure for each pair of officers. We adjust for crowd-out by down weighting by the number of other officers present. This process generates a time-varying network of potential contact between all officers in the organization. We then identify the officers likely to have received overtime from the wheel by comparing the seniority of each officer to the approximate position of the wheel on each date. An officer’s potential overtime supplier count on each date is then the sum of their potential contact with all other officers, weighted by each other officer’s proximity to the wheel on that date.

We present our results in five parts. First, we establish a set of stylized facts which justify our modeling choices and reinforce the claims from government documents. Because officers

are ordered on the wheel based on their seniority, and we observe hire dates, we are able to approximate the daily position of the wheel. Specifically, by taking the circular median of the seniority of officers that work on each date, we can observe the wheel turning in the data. The wheel turns often—as much as every day during the busy season—and when an officer is close to the median on a given date, they are more likely to work on that date. We show that despite the wheel’s operation, the top 10% of officers work 36.3% of all overtime shifts. We finally show that officers well-connected to those close to the wheel position on a given date (potential overtime suppliers) are much more likely to work overtime on that date. Specifically, an officer-date with a potential supplier count in the bottom quartile has an average overtime probability of 9.5%, while an officer with a potential supplier count in the top quartile has an average overtime probability of 25.4%.

Second, we estimate officer preferences and access costs. We exploit the long panel structure of our data and use a logistic regression estimator with both date and officer fixed effects. We can include high-dimensional fixed effects in a nonlinear model because the LADOT employs over 400 non-probationary traffic officers, and we observe 1.5 years of daily overtime decisions, allowing us to use asymptotic incidental parameter bias corrections. We capture time-varying access costs by including an officer’s angular distance from the wheel in terms of seniority on each date as well as an officer’s potential supplier count. We estimate that access costs are \$35.31 per shift, and an officer can reduce these costs by \$0.88 per potential supplier. We validate our model by using external data to show that our model estimates correctly capture the amenity value of different types of shifts.

Third, we perform a Shapley-style decomposition of the share of overtime worked by the top 10% of officers, our preferred measure of inequality. We shut down three dimensions of heterogeneity in our decomposition: informal networks (insider influence), wage differences generated by being on different pay scale steps, and officer time-invariant overtime preferences. Removing informal network heterogeneity increases the share of overtime worked by the top 10% of officers by 0.1 percentage points, which is small relative to baseline. In contrast, removing preference heterogeneity reduces the share of overtime worked by the top 10% of officers by 3.8 percentage points and removing wage heterogeneity reduces the share by 2.3 percentage points. Thus preferences and wage differences, not insider access, are the main drivers of observed inequality.

Fourth, we compare informal trade to random assignment, a counterfactual uniform-wage shift auction, where officers bid wages for overtime, and a counterfactual uniform-markdown shift auction, where officers bid markdowns from the existing pay scale. By construction, uniform-wage auctions assign shifts to the officers that value them most and therefore represents an upper bound on achievable allocative efficiency. Informal trade achieves 93.6% of

the gap between random assignment and the uniform-wage auction, which is \$4.32 million in earnings equivalent terms over a 1.5 year period. Informal trade is also more equitable than a uniform-wage auction and approximately the same as the uniform-markdown auction in terms of equity and efficiency. The officers who were likely labeled the cartel in the organization because they earned close to or above six figures in overtime continue to earn similar overtime income under the alternative centralized systems, and the wage savings from moving from informal trade to a uniform-wage auction are a modest 2.14%.

Finally, we ask why informal trade within the LADOT makes efficient allocations even when it is compared to relatively extreme centralized benchmarks. We vary overtime access costs and informal network mitigation up and down relative to our estimates, and change the correlation of officer centrality in the network with preferences, recomputing the allocative efficiency under informal trade. The primary reason that informal trade is efficient is that individual heterogeneity in preferences is an order of magnitude larger than access costs. Intuitively, officers that want work can and do obtain it. However, even if we scaled up access costs, the network remains relatively efficient. We show that this is because connectedness in the informal network is positively correlated with the preference to work. Intuitively, officers gain potential suppliers by working more in the past, and thus the network encodes the overtime preferences of officers, allowing it to make efficient allocations. The LADOT therefore illustrates a general idea: when trading networks are formed through the consumption of the traded good, a version of the Coase Theorem can be made to hold even in the presence of transaction costs, because the network lowers transaction costs for precisely the people who need to trade.

Our paper primarily contributes to the literature on the overtime compensation of public sector workers. Chalfin and Gonçalves 2023 study how the arrest decisions of police officers interact with the personal cost and benefits of overtime work, finding that while officers make fewer arrests near the end of a shift, these arrests are of higher quality. Similar to our paper, Chalfin and Gonçalves 2023 shows the existence of a potentially distorting force (overtime aversion in their paper, informal network trade of work in ours) that ends up having rather muted efficiency and equity implications in the aggregate. Bolotnyy and Emanuel 2022 study bus and train operators and show there is a large gender gap in earnings that can be attributed to differences in overtime and unpaid time off. Our results complement these findings, and consider how an informal secondary market for overtime layered on top of an ex-ante equal system shapes overtime earnings.

Our paper also adds to the related literature on the impact of coworker networks in another organization responsible for public safety and enforcing laws: police agencies. Several papers have studied the impact that an injury and death among police officers has on con-

nected officers. Holz, Rivera, and Ba 2023 considers the effect of a peer death on use of force, while Cho, Gonçalves, and Weisburst 2024 analyzes the effect of a peer injury or death on arrest rates. Other work considered more explicit peer networks such as police unions, and their potential effect on outcomes such as misconduct (Gonçalves 2021). As in our paper, these studies find that coworker networks can exert a heavy influence between public sector employees, even when those workers are not colocated during their shifts.¹

Our paper contributes to the literature on worker assignment within organizations by focusing on informal decentralized mechanisms and their impact on worker surplus. Past work has studied a variety of settings, including the IMF (Barron and Vardy 2005), Teach for America (Jonathan M.V. Davis n.d.), the U.S. army (Jonathan MV Davis, Greenberg, and Jones n.d.), and a large multinational corporation (Minni 2024), but has largely focused on formal centralized mechanisms like deferred acceptance or direct assignment by managers. Ba et al. 2022, which studies the impact of switching police officers from a seniority-based assignment system to alternative mechanisms, explicitly accounts for officer preferences and is most similar to our paper in that regard.

Our paper speaks to a longstanding literature on the relationship between firms and markets (Coase 1993, Rosen 1988). A motivating question in this literature is why organizations exist within markets. We document a case where a market exists within a rigid and unionized government organization. An important aspect of our setting is that social networks alleviate frictions in this informal market for overtime, and in that respect our paper is similar to Bandiera, Barankay, and Rasul 2012, who study how performance pay interacts with social networks within the firm.

Several papers show that the preferences of workers need not align with the outcomes desired by the organization employing them. Examples include software development workers trading off productivity and job satisfaction (Cowgill et al. 2024) and teachers disliking work at schools at which they have a comparative advantage (Laverde et al. 2023). We consider the organizational goals of minimizing overtime inequality and maximizing worker surplus, and show that the informal trading system used by traffic officers within the LADOT achieves these goals fairly well relative to alternatives. Our finding that uniform-wage auctions potentially worsen productivity is more in line with past work, in that it demonstrates a tension between worker preferences and a potential organizational goal.

¹Others have also focused on police agencies as workplaces. These include papers analyzing the effect on productivity of reference points in pay (Mas 2006), the effects of increasing police force size on crime reduction and discrimination (Chalfin, Hansen, et al. 2022), and the role of managers on team performance (Bacher-Hicks and Campa 2020).

2 Institutional Details

This paper studies traffic officers working in the Los Angeles Department of Transportation (LADOT). Los Angeles is divided into six geographic parking-enforcement districts. Each parking-enforcement district has an office. A map of offices and districts is provided in Appendix Figure A.1. Traffic officers operate over a large area: many parts of the Valley District are more than 40 miles from Harbor Traffic Control.

Traffic officers are allocated to divisions usually associated with the geographic districts. Regular work assignments are controlled at the division level, and special event work assignments, which we describe shortly, are disseminated to officers through their division but are largely coordinated centrally. Video footage posted by the LADOT shows that traffic officers do a roll call at their assigned office (LADOTOfficial 2018). For these reasons, we will refer to officers who work in the same division as “coworkers.”

Traffic officers have two core duties: parking enforcement and traffic control.² Parking enforcement duties primarily consist of patrolling a designated beat to identify and cite vehicles and drivers in violation of Los Angeles and California codes, while traffic control duties consist of directing traffic at assigned locations to prevent gridlock, placing traffic control devices (e.g., barricades or flares), and managing roadways when traffic accidents occur. While some traffic work is routine (e.g., daily management of an intersection known to suffer from congestion), the vast majority is caused by permitted *special events*, which include the Los Angeles Marathon, Dodger games, the Oscars, parades, protests, and block parties. Los Angeles hosts several thousand permitted special events every year. Whenever a special event would impact traffic and parking in the city, traffic officers are dispatched. Thus, while these special events can create additional labor demand for traffic officers, these events involve many of the same tasks as routine traffic control.

Around 80% of regularly scheduled traffic officer time is spent on parking enforcement, and most traffic officers are allocated exclusively to parking enforcement duties for their regularly scheduled working hours. Since the mid-2000s, the routine traffic control duties that comprise the other 20% of regularly scheduled time have also been performed by officers allocated exclusively to traffic control. Officers generally work alone when performing regularly scheduled parking-enforcement and traffic-control duties. Large special events are the primary scenario in which multiple officers would be working near each other.

We highlight that traffic officers are highly substitutable between shifts and work assignments. While the assigned beat and parking conditions can vary, the core task of parking

²City of Los Angeles Civil Service Commission 2009 describes the traffic officer role, and City of Los Angeles Personnel Department 2024 describes the core tasks.

enforcement is fundamentally identical between officers. While each special event is unique, traffic direction always consists of the same small set of tasks, several of which the LADOT documents in YouTube training videos (Parrish 2020). These tasks do not appear to create much opportunity for teamwork or complementarities between officers; a special event as large as the Oscars will require a large group of traffic officers, but there is not much scaling in the complexity of the work. Different traffic officers will have varying levels of productivity and skill but there is limited opportunity or need for specialization. This feature is highlighted in Appendix Figure A.2, which shows the relative simplicity and lack of specialization in the career ladder for traffic officers relative to firefighters in Los Angeles.

Traffic officers are union employees covered by the Memorandum of Understanding 18 (MOU) between the City of Los Angeles and Service Employees International Union Local 721 (City of Los Angeles, Office of the City Administrative Officer 2015). According to the MOU, traffic officers are assigned fixed and regular workweeks and work schedules (e.g., a 5/8 or a 4/10 work schedule). The regular salary for traffic officers is standardized by the MOU. During the period of our analysis, the MOU establishes the overall salary range as roughly \$47,000 to \$64,000. Exact salary is determined by tenure using a step structure. After completing the probationary period, a traffic officer will generally qualify for a step increase every twelve months, translating into a roughly 3% annual raise.

A consequence of the high degree of substitutability is that most traffic officers are eligible to complete overtime work, should the need arise. Based on a conversation we had with a traffic supervisor that we confirmed in our data,³ the only officers excluded from special events overtime are those in their probationary period. Non-probationary Los Angeles traffic officers control their labor supply by working overtime shifts. Over 150,867 hours were billed to overtime pay codes in calendar year 2015. This overtime comes from two sources. First, routine overtime arises from excess demand for traffic control due to something like an emergency (e.g., a burst pipe or a broken traffic light) or a construction project. Second, pursuant to the MOU, all special events work is paid out as overtime. Based on city reports, in fiscal year 2013-2014 (the year prior to our data), 80% of all overtime paid out to employees within the Department of Transportation Parking Enforcement and Traffic Control was due to special events.

The MOU prescribes a general policy for the assignment of overtime among traffic officers: “Management will attempt to assign overtime work as equitably as possible among all qualified employees in the same classification, in the same organizational unit and work

³Probationary officers appear in our data and in city documentation with the job title “Traffic Officer I,” while non-probationary officers have the Title “Traffic Officer II.” Across 43,805 person-days in our period of analysis, we find only one record of a Traffic Officer I working an overtime shift.

location” (City of Los Angeles, Office of the City Administrative Officer 2015, 27). How exactly overtime is assigned to individual officers is not spelled out in the MOU. However, a report by the City Controller’s office (Galperin 2015) gives more detail about special-events overtime. Any event with sufficient advance notice is coordinated centrally by LADOT, and overtime is assigned using a mechanism officers call “spinning the wheel.” Under the wheel system, officers first volunteer to be on an overtime list. The list is ordered by seniority. At the start of each month, the DOT “spins the wheel” by assigning special-event overtime sequentially to officers in the order they appear on the list. The list is called a wheel because once the list is exhausted, it cycles back to the beginning again. Thus seniority serves as reference for where an officer is on the wheel, like the hours on a clock, but does not determine how much overtime an officer receives over a long horizon. For those events with less than six days’ advance notice, an analogous version of the wheel is spun at local area offices, using a list containing only the officers assigned to that office. The wheel system is implemented via a Microsoft Access program, created as part of a traffic officer’s master’s thesis.

The LADOT continues to spin the wheel month after month, as do the district offices with their wheel. Because the wheel spins continually, and given that thousands of special events occur each year in Los Angeles, the initial assignment of any particular officer to any particular special event is as good as random, and the cumulative amount of overtime assigned over any period longer than a few weeks will be similar across any officers who have volunteered. The city report states this explicitly:

The wheel system ensures that overtime is initially assigned equally to all employees who desire to work overtime. Over a given time period, each employee in the Access database will basically be assigned the same number of special events.

In practice, the observed amount of overtime between officers varies wildly, a fact we document in Figure 3. This imbalance in overtime hours is largely the result of shift trading between officers. Once the wheel has been spun, special-event assignments are sent to area supervisors and posted on bulletin boards in staff areas. After seeing their assignment, an officer assigned an overtime shift may work it or find a substitute. Documentation of substitutions is not enforced by LADOT, but the aforementioned City Controller’s report recounts that such trading is common. While only 192 officers signed up to volunteer in FY 2013-2014,⁴ 471 officers worked overtime, suggesting that not only is trading pervasive, it even induces non-volunteers to work special events. The wheel is not adjusted to accommodate the normal schedule of officers. Instead, during the period we consider, officers had wide latitude to use accumulated sick time or leave time call out of their normally scheduled work

⁴This is the year prior to our analysis period, which spans part of FY 2014-2015 and all of FY 2015-2016.

in order to work a special-event overtime shift instead.⁵ In sum, the combination of written policies and departmental norms within LADOT created a workplace in which most officers could work most overtime shifts, provided they could find a trading partner.

A natural question is why an officer would choose not to volunteer to be on the wheel. After all, an officer could simply give away their shift if they do not want to work and potentially obtain favors or cash from someone who wants to work. We discussed this with a supervisor within the LADOT, who told us that an officer can face consequences if they are assigned special events by the wheel and no one shows up to work their slot. One such consequence is removal from the wheel. Thus, if there are trading frictions or some specific special events that are not popular to work, being on the wheel is not costless.

Crucially, trading overtime shifts requires a bilateral trade: an officer must give up an assigned shift and another officer must take on an unassigned shift. Special-event overtime guarantees a higher pay rate and also the amenity value of working certain high-profile events and so many officers might be willing to trade into an overtime shift if it is not costly to do so. Among the thousands of special events in the period, some are particularly likely to involve exposure to celebrities. These include public-facing events like the Oscars as well as private celebrity parties. Notable examples are provided in Appendix Table A.1.

Shift trading between traffic officers may involve banking favors to be repaid in the future, the use of social status, and even side payments in cash.⁶ We remain agnostic about the exact currencies used to facilitate any particular trade, but highlight that these trades all happen within a network. After the wheel is spun, special-event overtime assignments are publicly posted on bulletin boards in division offices. An officer seeking to trade for a desirable shift or trade away an undesirable shift needs to identify trading partners and convince them to trade. According to a supervisor, trades are facilitated both via word-of-mouth and via an informal online forum. In theory, any non-probationary officer may trade a shift; however, in practice an officer who cultivates a more extensive or dense network should have an advantage among their peers with regard to the flexibility and availability of overtime shifts. In city reports, Parking Enforcement and Traffic Control staff have raised concerns about officers who use their networks in exactly this way. Referring to these officers as “The Cartel,” Galperin 2015 highlights that officers cultivate relationships to obtain high-value

⁵This possibility was in fact intended by the LADOT and union to create more equitable access to overtime: unless officers were allowed to call out of a normal shift and work a contemporaneous special event, officers with work schedules that substantially overlap special events would rarely be able to work overtime.

⁶There is some disagreement between Galperin 2015 and a supervisor we spoke to within the LADOT about whether cash is used. Anecdotal reports from other government organizations suggest side payments do occur elsewhere.

special-event overtime shifts.

Even if an officer cannot arrange an official trade, it is possible to work a special event by essentially paying a waiting cost. In conversations with a supervisor, we learned that officers will sometimes line up at the meeting point for a special event and take the slots of officers assigned the event who do not show up. This is possible because special event assignments with meeting locations are posted on public bulletin boards.

3 Data

We combine data from several sources. Our primary data set consists of payroll records for all Los Angeles traffic officers. The payroll data record daily work activity, including amount and type of wages paid and hours worked. The data come from a discontinued city project, and it was subsequently released as public record.⁷

We construct a daily panel data set tracking overtime and standard work between July 1, 2014 through June 30, 2016. We use the first six months (July 2014 through December 2014) to construct an initial history of officer work patterns. The rest of the data, which starts on January 1, 2015 and ends on June 30, 2016, is our analysis period. Summary statistics are provided at the officer-date level in the top panel of Table 1, and at the officer-level in the bottom panel. For officer-level statistics, we compute tenure and age to be as of January 1, 2015. We report summary statistics of number of regular shifts worked, number of overtime shifts worked, and number of dates active in the panel.

Table 1: Summary Statistics

Variable	Mean	SD	P25	P75
<i>Officer-Day Summary Statistics (N officers = 535, N observations = 274,908)</i>				
Wage	29.988	2.356	30.540	30.540
Age	46.792	8.984	40.289	53.368
Tenure	15.039	8.140	9.139	19.313
Standard Work	0.673	0.469	0.000	1.000
Overtime	0.139	0.346	0.000	0.000
Expected Earnings	283.076	61.282	240.133	317.489
<i>Officer-Level Summary Statistics (N officers = 535, N observations = 535)</i>				
Wage	29.500	2.838	30.470	30.472
Age	45.843	9.198	39.254	52.761
Tenure	14.155	8.482	8.192	18.563
Days Active	513.847	95.414	547.000	547.000
Standard Work	345.817	65.884	368.000	372.000
Overtime	71.258	86.013	4.000	121.000

Officers vary significantly in age, but many are in their 40's with more than 8 years of

⁷We provide the raw data in full at <https://www.jkohlhepp.com/publication/selection-injury/>.

traffic officer experience. Wages are quite similar across officers, reflecting a compressed, union-negotiated pay scale. We report expected earnings for each officer-date, which we compute as the average number of overtime hours among those who work overtime on each date multiplied by an officer’s overtime rate. Most officers are active and observed for all 547 days of the analysis period. Most officers are observed working non-overtime (standard-work) hours on 370 days, with 50 percent of officers working between 368 and 372 non-overtime days. Throughout the paper, we refer to an overtime shift as a day when an officer is observed working positive overtime hours. In contrast, the number of days each officer works overtime varies substantially, a fact that we document more extensively in the next section.

We augment the payroll records with information about daily weather patterns obtained from National Oceanic and Atmospheric Administration [2019](#). We obtain special event permit information from the dataset Los Angeles Department of Building and Safety [2025](#) which is publicly available on the Los Angeles City Data Portal.

4 Stylized Facts

We present a series of stylized facts which show that transaction costs exist, significant overtime trade occurs despite these transaction costs, and informal networks created through past contact play a key role.

An important feature of our setting is that conditional on volunteering, being assigned overtime on a particular day is completely determined by the mechanical turning of the wheel. Further, an officer’s position on the wheel is entirely based on their seniority rank within the organization. The Coase Theorem states that without transaction costs, efficient trade should occur regardless of how properties rights are initially allocated. Thus, if the wheel can be observed turning among the set of officers that end up actually working the shift, this is evidence that transaction costs exist. To operationalize this, we compute the seniority rank of officers on each date based on the set of active traffic officers. We then convert each officer’s seniority to an angle on the unit circle, where the least senior officer on a date is 360 degrees and the most senior is $360/N$. We then compute the circular median on each date among only officers that work overtime, and obtain the following result.⁸

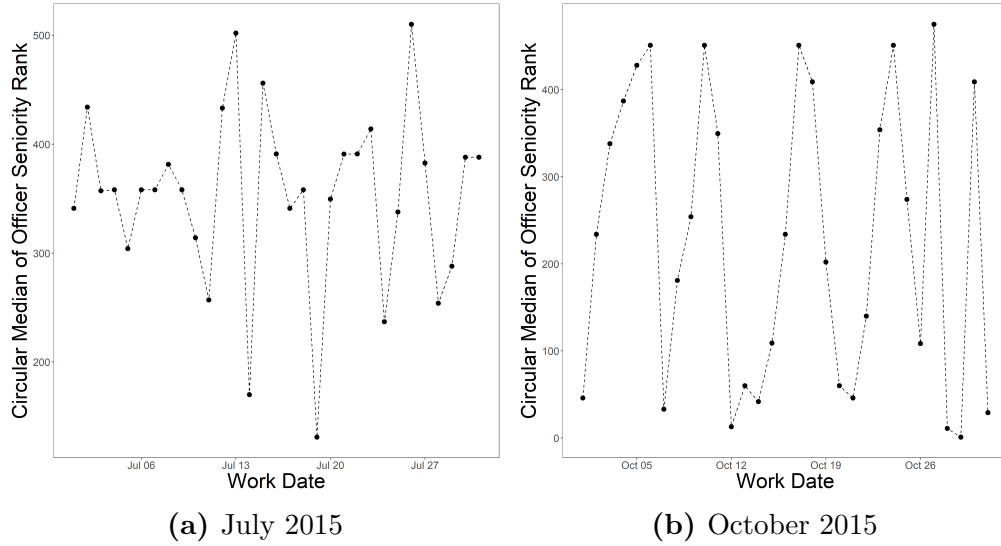
Fact 1. The circular median of seniority rank among officers that work overtime follows a cyclical pattern across time.

Put another way, the wheel can be observed turning in the aggregate overtime work patterns. We illustrate the result graphically for July and October 2015 in Figure [1](#). Similar

⁸The circular median accounts for the fact that the most junior officer is close to the most senior officer.

patterns arise in other months and when the circular mean is used rather than the median. In October, the revolutions of the wheel can be clearly observed: the circular median of seniority begins close to 1 (the top of the list), rises above 400, then falls back down again, completing around five cycles in the month. The latter half of July looks similar to October, but the early part of July features a period where the circular median stays roughly stable for 1.5 weeks. This is because early July is one of the busiest times for special events (and therefore overtime), and as a result the wheel is making a full rotation almost every day.

Figure 1: The Overtime Wheel Turning



We stress that although we can approximate the position of the main organization-wide wheel on any given date, exact reconstruction of the wheel is not possible given the available data.⁹ Nevertheless, we can show that an officer's distance from the approximate median of the wheel on any given date impacts the probability an officer works on a given date.

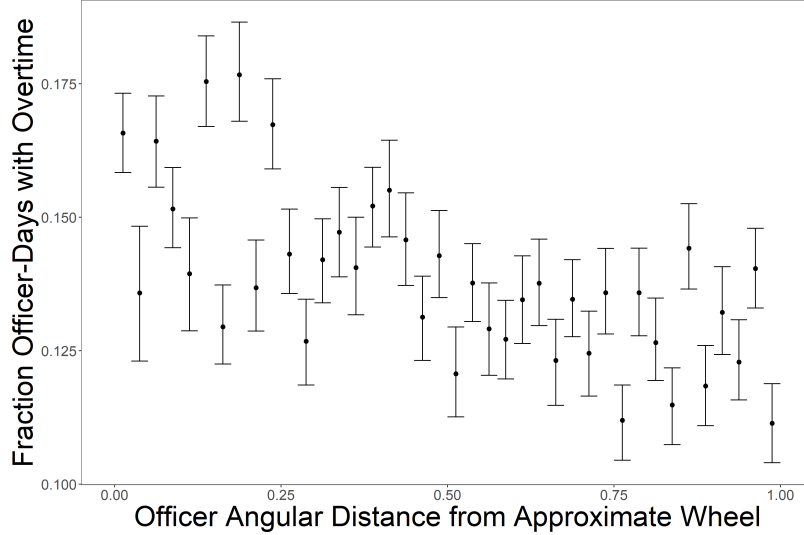
Fact 2. An officer with a seniority closer in angular distance from the approximate median of the wheel is more likely to work overtime.

When computing the angular distance on date t of an officer i , we leave officer i out of the circular median of the seniority of officers who work overtime on date t . We plot the fact visually in Figure 2 by computing the overtime probability among officer-dates in 40 equally spaced bins based on angular distance from the leave-self-out median. The maximum distance of 1 corresponds to being on the other side (180 degrees) of the observed median of officers working that day, while a minimum distance of 0 corresponds to falling exactly on

⁹This is because multiple smaller wheels also operate, officers volunteer to be on the wheel, and around 20% of overtime does not come from the wheel.

the observed median.

Figure 2: Distance from the Wheel and Overtime Access



Note. Standard errors are computed using 1,000 nonparametric bootstrap replications. The approximate wheel position is the leave-self-out-median of all officers that worked on each date.

When an officer is almost exactly on our estimate of the approximate middle of the wheel, their probability of working overtime is 0.17. When an officer is on the opposite side of the wheel, their probability drops towards 0.12. The last two facts confirm the important role of the wheel in assigning overtime, and they illustrate that there exist transaction costs within the organization that prevent overtime from flowing to the person who values it most on each date.

However, if transaction costs are large, trade would not occur at all. Because the wheel turns several times or more a month and our data spans more than a year, this would further imply an even distribution of overtime among volunteers. We emphasize this point: although seniority determines an officer's position on the wheel, because the wheel never resets more senior officers are assigned at most one more overtime shift in any given period of time, which means that in the long-run seniority does not confer an advantage in obtaining overtime from the wheel. Practically, these facts imply that absent trading, we should observe a bimodal distribution with a large mass of non-volunteers working close to zero overtime and a large mass of volunteers working the same amount of overtime. This is not what we observe.

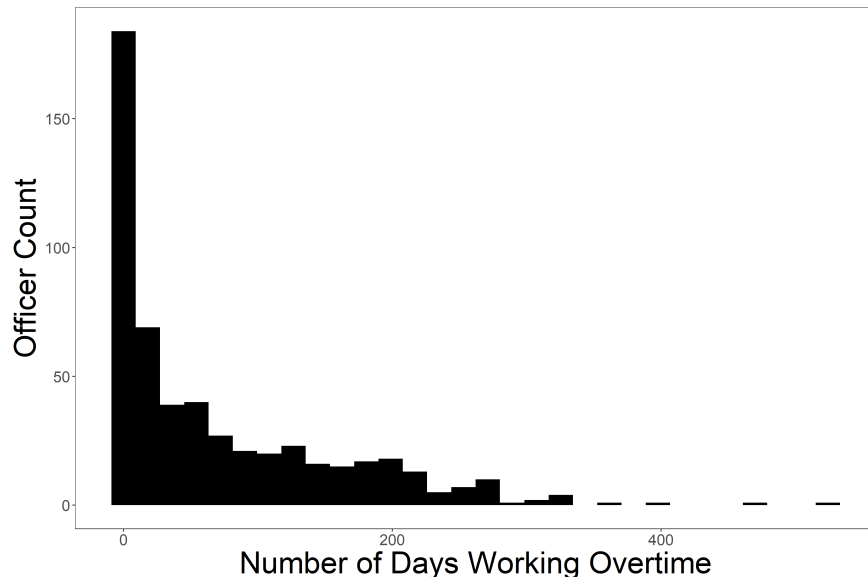
Fact 3. A small number of officers work a disproportionate share of overtime.

We present Fact 3 visually in Figure 3, which plots a histogram of officers by the number of distinct dates they are observed working overtime. As hypothesized, we observe a large mass of officers, likely non-volunteers, working little to no overtime. But instead of a second mass of officers working a similar and positive amount of overtime, we observe a long-right

tail, with a small number of officers working a disproportionate share of overtime. In the very far right tail, we confirm the existence of some officers who might be part of the “cartel” documented in the government report. These officers are observed working overtime on 300 or more distinct dates in a 546-day period. In terms of inequality, the top 10% of officers work 36.3% of all overtime in the analysis period.

We provide several statistics to quantify the dispersion in overtime. An officer at the 25th percentile works 4 overtime shifts, while an officer at the 75th percentile works 121 shifts, which is 30.3 times more. If we limit only to officers observed working at least one overtime shift, dispersion is still large. An officer in this subsample at the 25th percentile works 12 overtime shifts, while an officer at the 75th percentile works 134 shifts, which is 11.2 times more.

Figure 3: Distribution of Overtime across Officers



The skewed distribution of overtime is evidence that significant trade occurs despite the existence of transaction costs. What facilitates trade absent a centralized or formal market? The government report provides an answer: “Traffic Officers may be able to receive more overtime if they have nurtured relationships and know how to network, treating overtime assignments as a privilege that can be traded.” In other words, informal coworker networks within the organization are used to trade overtime. To assess this claim, we measure how well connected each officer could be within the organization, given work patterns in the time leading up to each overtime opportunity, and then we combine this information with the approximate position of the wheel.

We start by constructing the potential contact between two pairs of officers on date t ,

which we denote $p_t(i, i')$. For officer pair (i, i') on date t , we check whether both officers worked at the same division. If they did not, we set $p_t(i, i') = 0$. If they did overlap, we compute the total number of officers who worked at that division, call it $N_{i,t}$ and set $p_t(i, i') = \frac{1}{N_{i,t}-1}$. We measure the total potential contact between officer pair (i, i') prior to date t in the past $M = 90$ days as $c_t(i, i') := \sum_{\tau=t-1}^{t-M} p_\tau(i, i')$. We compute this measure for all pairs of officers and for all dates.¹⁰

The active officers on any date t are those hired on or before t and those terminated on or after t . For any fixed t , we can use the set of active officers and their pairwise measures $c_t(i, i')$ to construct a weighted undirected network G_t , which we call the network of potential contact. Two officers who never worked in the same division on the same day in the last 90 days are unconnected because $c_t(i, i') = 0$. Among those with a connection, the strength of that connection is proportional to the number of times they worked alongside each other weighted by the number of other officers present. Weighting by the inverse of the number of other officers present is meant to adjust for crowd-out: officers who overlap on a day where many other officers are working have less one-on-one contact. For example, parking rules are relaxed on Sundays in Los Angeles, and the data reflect that fewer traffic officers work on Sundays. If two officers do happen to work on Sunday, we will calculate a high value of $c_t(i, i')$ because there are fewer other officers to crowd out their interaction.

The measure $c_t(i, i')$ is meant to capture the fact that officers likely report to a shared district office to pick up their vehicles and gather information, but leave the office to perform their duties on different routes throughout the district. We provide two pieces of evidence that support this view. First, promotion videos posted by the LADOT on YouTube show officers gathered within one parking enforcement district Office, engaging in what appears to be a daily roll-call meeting (LADOTOfficial 2018). Second, the initial assignments procedure for each specific special event are printed on physical sheets of paper and posted on a bulletin board in each division office. According to a source within the department, the board is updated multiple times a day. We provide an actual picture of an initial assignment from 2025 as well as one of the bulletin boards in Appendix Figure A.3. In order to effectively trade, an officer needs to both know their own initial allocation and the allocations of other officers, which requires physically checking the board. This would be a natural thing to do while working, and it would also be a natural place to meet potential trading partners.

We define a measure which summarizes an officer’s centrality within the network of potential contact.¹¹

¹⁰We also compute the measure for different rolling window lengths M .

¹¹Our measure corresponds to the number of direct links. We considered incorporating indirect links

Definition 1 *Officer i 's connectedness on date t is denoted by $c_{i,t}$, and is given by the sum of their potential contact with all other active officers up to but not including date t :*

$$c_{i,t} = \sum_{i' \neq i} c_{t-1}(i, i').$$

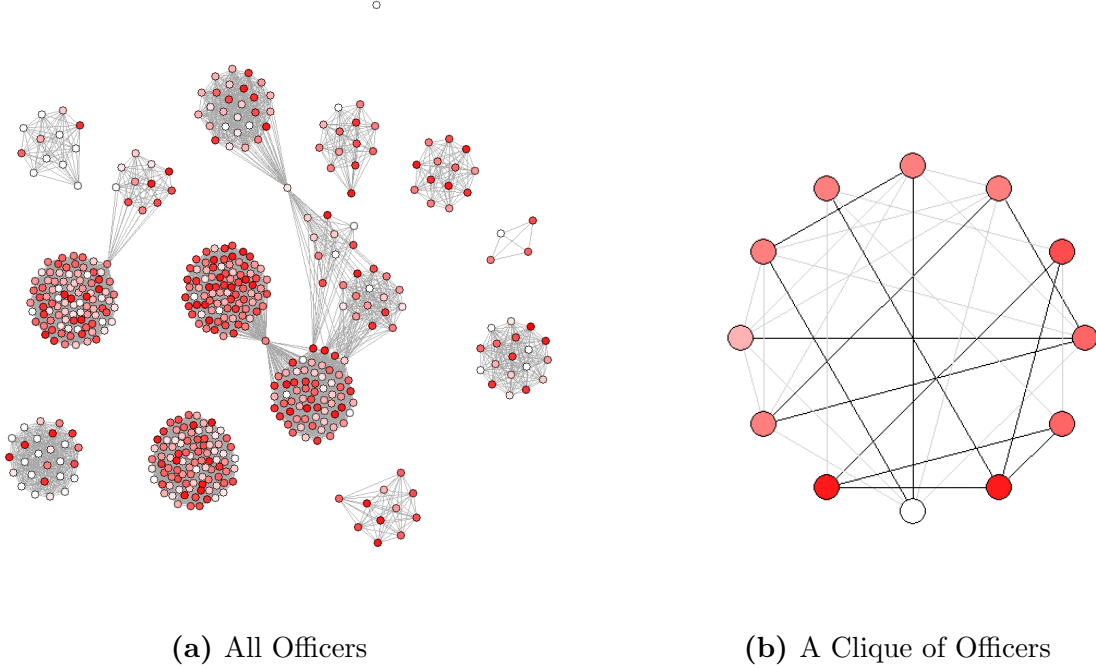
A one-unit increase in the measure is equivalent to one additional opportunity for a one-on-one interaction with another officer. A half-unit increase is equivalent to one additional opportunity to interact with another officer when a “third-wheel” officer is present. Connectedness measures how well-connected the officer could be given the set of officers they worked alongside prior to date t , incorporating both the breadth and depth of connections. Consider a connectedness of 60, a common value in the data. An officer could have this value if they overlapped with the same other officer in the same location for 60 of the last 90 days, and had no potential contact with anyone else. An officer could also have this value because they overlapped with two other officers in different locations for 30 days each. We illustrate how connectedness varies in the cross-section by depicting the network of potential contact on the first day of the analysis period in Figure 4. Each node is an officer, and each officer is colored based on the quantile of their connectedness on January 1, 2015. In Panel A we show the full network, where officers with any potential contact in the last three months are connected by a line. Large clusters typically reflect officers working in the same geographic division.

In Panel B we zoom in on a small disconnected part of the network to show that connection strength varies even within groups of officers who have all worked together. Schedule differences and variation in how often officers worked in the past drive differences in the strength of pairwise connections. All the officers in Panel B have worked together in the last 3 months, but a group within a group with especially strong connections (5 or more pairwise interactions). In Appendix Figure A.4, we illustrate within officer across time variation in officer 230's direct connections. The composition, number, and relative strength of officer 230's direct connections change throughout the period. This occurs both for endogenous reasons (an officer who chooses to work more overtime will be more connected) but also plausibly exogenous reasons (a coworker may retire or switch divisions).

For an officer to obtain overtime on the secondary market, they must not just know people; they must also know the right people, more specifically the people endowed with overtime by the wheel. Thus, the wheel system creates plausibly exogenous variation in access in both the cross section and within officer across time. To measure this variation,

using a different centrality measure but found this measure was less positively correlated.

Figure 4: The Network of Potential Contact on January 1, 2015



Note. Panel A features all employed traffic officer IIs on January 1, 2015. Two officers are plotted as connected if there is strictly positive potential contact in the last 3 months. Panel B focuses on a set of officers who are all connected to each other but disconnected from all others. Connections are plotted with a darkness based on the amount of potential contact. Officer node color is based on the connectedness, with red denoting high connectedness.

we compute the angular distance of officer i 's seniority from the circular median seniority among all officers who worked on date t . We exclude officer i from the circular median if they happened to work overtime that day. We call this quantity $A_{i,t}$, and we measure it in terms of half rotations, so that it is 1 when an officer is maximally distant from the approximate median of the wheel. We then use it to construct the following measure.

Definition 2 *Officer i 's potential supplier count on date t is denoted by $z_{i,t}$, and is given by the sum of their potential contact with all other active officers up to but not including date t , weighted by 1 minus the distance of each other officer from the leave-self-out circular median of the wheel on date t :*

$$z_{i,t} = \sum_{i' \neq i} (1 - A_{i',t}) c_{t-1}(i, i').$$

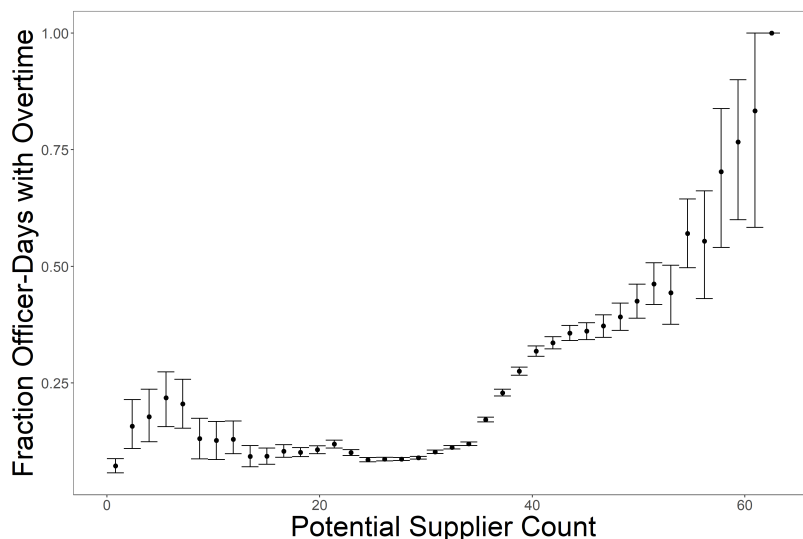
An officer's potential supplier count is distinct from their connectedness because it accounts for the fact that a connection should matter for obtaining overtime on date t only if the connection happens to be endowed with overtime by the wheel on date t . Thus a well-connected officer who has many connections far from the wheel position on a given

date can have fewer potential overtime suppliers than a less well connected officer who has a handful of connections close to the wheel’s position. As a result, there is much greater variation in potential supplier count than connectedness, as shown in Appendix Figure A.5. The distribution of potential supplier count has a shape similar to a normal distribution, and most of the variation comes across time within officer: a regression of potential supplier count on officer fixed effects accounts for only 28.4% of the variation.

Fact 4. There is a strong positive association between potential supplier count and the probability of working overtime.

We capture this fact visually in Figure 5, where we compute the fraction of officer-days with overtime within 40 bins with an equal number of observations. An officer-date in the bottom quartile of potential supplier count has an overtime probability of 0.095, while an officer-date in the top quartile has an overtime probability of 0.254. A potential supplier count near 60 is rare because it requires both high connectedness and alignment of the wheel, but when it happens the officer in question is observed working overtime the vast majority of the time. Figure 6 shows the strong positive association persists when we residualize both overtime and potential supplier count for officer fixed effects, date fixed effects, and several other controls.

Figure 5: Potential Supplier Count and Overtime Access



Note. Standard errors are computed using 1,000 nonparametric bootstrap replications.

Fact 4 is consistent with both the government report and conversations with a supervisor within the LADOT. The wheel determines initial allocations of overtime fairly. Afterwards, officers can bilaterally buy, sell or trade overtime without management intervention. This exchange is done by word of mouth as well as via informal online forums. As the government

report suggests, well-connected officers can obtain significantly more overtime because of better access. If officers all had the same preferences for overtime, we could conclude from Fact 4 that a greater number of suppliers reduce the cost of obtaining overtime, either by reducing transaction costs or by improving the buyer’s bargaining position. In the next section we present a model we will use to disentangle preferences from access costs.

5 Empirical Strategy

We introduce a simple model where officers decide each day whether to work an overtime shift. The goal of the model is to separate preferences from access costs. We capture time-varying access costs due to direct assignment by the wheel using an officer’s angular distance from the wheel position on each date. We capture time-varying access costs due to the secondary market using the potential supplier count.

5.1 Model

There are N officers indexed by i . An officer makes myopic daily decisions to work an overtime shift on dates $t = 1, 2, \dots, T$. Officer i ’s utility from working an overtime shift on date t is given by

$$u_{i,t} = \rho Wage_{i,t} + X'_{i,t} \beta_1 + \gamma_t + \alpha_i + \epsilon_{i,t}.$$

The monetary value of an overtime shift is captured by $Wage_{i,t}$, which is each officer’s base rate times 1.5 (the overtime premium). Base rates mainly vary mechanically based on tenure and the union-negotiated pay scale. Officer-specific time-invariant utility from overtime is represented by α_i . Date-specific amenities of overtime are represented by γ_t . Utility is also allowed to depend on seniority rank within the organization on date t and whether the officer works standard hours that day, which are both included in $X_{i,t}$. We include seniority rank in utility in case there are effects of seniority not captured by our specification of access costs. We capture idiosyncratic taste for overtime with $\epsilon_{i,t}$, which we assume is i.i.d. standard logistic.¹² The parameter ρ converts non-wage amenities into the wage shift that generates the same utility. We define the earnings equivalent of utility x as x/ρ multiplied by 6.49, the average overtime shift length during the period.¹³

¹²This is equivalent to assuming a type-1 extreme value structural error term in the utility from overtime and no overtime.

¹³We use earnings equivalents primarily to compare wage costs and worker surplus.

The cost of obtaining overtime is given by $A_{i,t}(\beta_2 + \beta_3 Z_{i,t})$, where recall $A_{i,t}$ is an officer's angular distance from our estimate of the median wheel position on date t and $Z_{i,t}$ is the potential supplier count of officer i on date t . Under this formulation, an officer exactly at the median faces no access costs because $A_{i,t} = 0$. Thus our specification embeds the assumption that on average, working an overtime shift when assigned directly by the wheel is costless. Given Fact 4, we expect more potential suppliers to reduce access costs, that is $\beta_3 > 0$. When an officer has no potential suppliers and is maximally distant from the wheel, ($Z_{i,t} = 0, A_{i,t} = 1$) access costs are at their maximum and given by β_2 . The reduction in access costs that comes from one additional potential contact being exactly at the median of the wheel is given by β_3 .

We denote the binary overtime decision $W_{i,t}$. We normalize the utility of not working overtime to be 0. Officers make overtime decisions to maximize utility less costs, which implies their observed overtime decisions can be expressed as

$$W_{it} = \begin{cases} 1 & \text{if } \rho Wage_{i,t} + X'_{i,t}\beta_1 + \gamma_t + \alpha_i + \epsilon_{i,t} + A_{i,t}(\beta_2 + \beta_3 Z_{i,t}) \geq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The single-threshold model implied by this specification of utility and costs reveals the key empirical challenge. An officer who does not work many overtime shifts may be doing so either because they prefer not to work overtime or because access is costly given their wheel position and their potential suppliers within the organization. Disentangling preferences from access requires accounting for the officer's position in the organization on each date. If this is not done, our estimates of the main source of preference heterogeneity, $\hat{\alpha}_i$, will be biased.

Our preferred specification does not include dynamic terms, like lagged overtime, which would allow for diminishing marginal utility from overtime. In Appendix Section ??, we estimate models with lags of both overtime and normal work and find positive coefficients on all lags of overtime, suggesting that there is no evidence of diminishing marginal utility from overtime. Because the wheel cycles frequently, the coefficients from such a lagged specification should be interpreted with caution. We provide a discussion of how peer personnel changes map into own changes in connectedness in Appendix Section B.

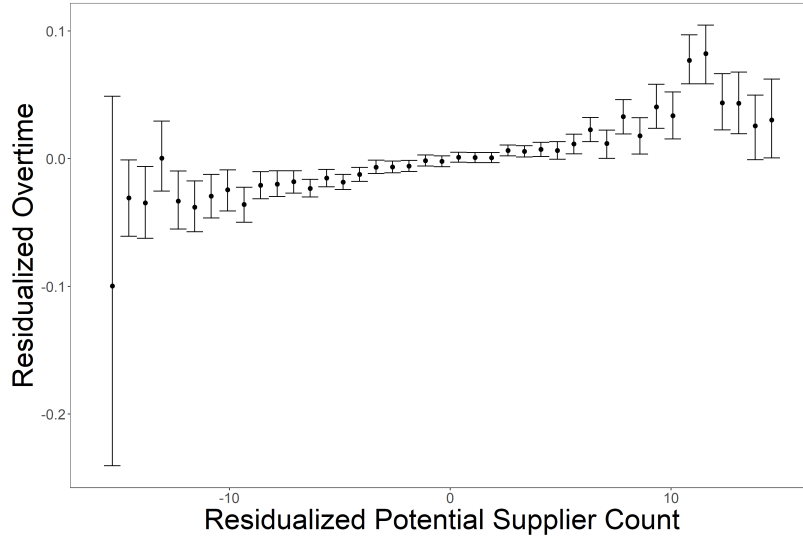
5.2 Testing Identifying Assumptions

Identifying officer preferences relies on two main assumptions. First, angular distance from the wheel and potential supplier count must capture the cost of obtaining overtime. The government report states that networking and connections are drivers of overtime disper-

sion. Because we do not observe trades, we cannot test this directly. But we can test it indirectly: if potential supplier count captures the cost of obtaining overtime, it should be relevant to the decision to work overtime even conditional on all control variables, most importantly officer fixed effects. Stated more formally, $Cov(W_{i,t}, Z_{i,t}|i, X_{i,t}, Wage_{i,t}, A_{i,t}) > 0$. Second, within-officer across time variation in potential supplier count cannot be correlated with within-officer across time variation in officer preferences for overtime. Stated formally: $Cov(Z_{i,t}, \epsilon_{i,t}|i) = 0$.

We provide evidence that potential supplier count is relevant to the overtime decision conditional on all other controls in Figure 6.

Figure 6: Residualized Binned Scatter Plot of Potential Supplier Count and Overtime



Note. Standard errors are computed using the nonparametric bootstrap and 1,000 replications. Both variables are residualized for date and officer fixed effects as well as the other time-varying controls. Observations with residualized supplier count in the top 1% and bottom 1% are excluded for visual clarity.

Comparing Figure 6 to the unconditional binned scatter plot in Figure 5, we see that residualizing removes some of the association between potential supplier count and overtime. Most of this correlated variation is cross-sectional and is removed by the officer fixed effects, reflecting the fact that officers who enjoy working overtime are on average better connected within the LADOT. Still, residualized potential supplier count remains strongly positively correlated with working overtime: across its support, average overtime probability increases by around 10 percentage points.

We provide two pieces of evidence that $Cov(Z_{i,t}, \epsilon_{i,t}|i) = 0$. We first note that this is a conditional requirement: the individual fixed effects allow officers with a strong time-invariant preference for overtime to have more potential suppliers on average. Consistent estimation of Equation 1 requires only that within-officer across time variation in potential

supplier count is not correlated with within-officer across time variation in preferences for overtime. The assumption is also crucial for our interpretation of β_3 as the marginal reduction in the cost of obtaining overtime as an officer becomes more connected to another officer who is endowed with overtime. If $Cov(Z_{i,t}, \epsilon_{i,t}|i) > 0$, β_3 will implicitly include officer preferences and we will underestimate access costs. When we set β_3 to be 0 in our counterfactual exercises, we would then be accidentally modifying officer preferences rather than holding preferences fixed and modifying only the assignment system.

We first proxy for time-varying overtime preferences using family medical leave and bereavement pay codes. The argument underlying this test is as follows. The death of a family member, the birth of a child, or the need to take care of a family member all represent negative time-varying shocks to the value of working overtime ($\epsilon_{i,t}$). If such shocks are associated with reductions in potential supplier count, this is direct evidence that our assumption $Cov(Z_{i,t}, \epsilon_{i,t}|i) = 0$ is violated.

We implement this test by creating an indicator $D_{i,t}$ where an officer is treated when they are first observed with the relevant pay code, which is defined as one with the text “BEREAVEMENT” for the bereavement “treatment” and “FML” or “FAMILY” for the family medical leave “treatment.” We observe 57 officers “treated” with bereavement at some point in the sample and 424 “treated” by family medical leave at some point in the sample. The outcome is potential suppliers without any lag. We estimate the average treatment effect on the treated using the Sun and Abraham 2021 estimator as well as two-way fixed effects and report the results in Appendix Table 2. Note that the Sun and Abraham 2021 estimator required aggregating the data to be at the weekly level due to numerical and sample size problems.

Table 2: Own Personnel Changes on Potential Supplier Count

Specification	Estimator	ATT	Std. Error	p-value	Observations
FMLA	TWFE	0.566	0.274	0.040	274,908
	Sun Abraham	-0.795	0.356	0.026	39,759
Bereavement	TWFE	0.605	0.733	0.410	274,908
	Sun Abraham	0.213	0.195	0.277	39,759

Note: Standard errors are clustered at the officer level.

To further interpret these results, we provide event study plots for the Sun and Abraham 2021 estimates in Figure A.6.¹⁴ We find no evidence that bereavement reduces connectedness: the Sun and Abraham 2021 estimator is 0.213 and is not statistically significant at the 0.05 level. We find weak evidence that FMLA reduces connectedness: the Sun and Abraham

¹⁴See Appendix Figure A.11 for the two-way fixed effects event study plots.

2021 estimator is -0.795 and is statistically significant at the 0.05 level.

We view this second piece of evidence as weak for two reasons. First, it represents a small effect: the total impact of an officer taking FMLA on their potential supplier count is 0.121 standard deviations on average. Second, FMLA does not appear to generate immediate dynamic effects in the event study plot. Rather, there appears to be a small negative effect that is not statistically significant in any given week but that aggregated across all weeks after treatment leads to a statistically significant average treatment effect on the treated. Our strategy of including officer fixed effects for a 1.5 year period should be able to absorb much of this slow moving cumulative effect. Despite these qualifications, we view the correlation of FMLA with potential supplier count as a limitation: there are important dimensions of officer preferences that are time-varying that our analysis is not well equipped to quantify or address.

We second check to see how much of the correlation between connectedness and overtime is due to connections formed very recently. We focus on connectedness because potential supplier count varies dramatically day by day due to the mechanical turning of the wheel. The argument underlying this test is as follows. As an officer approaches date t , more and more of the time-varying component of overtime utility ($\epsilon_{i,t}$) becomes known. For example, an officer with children does not know whether he/she will be well-rested prior to the shift until the morning of the shift. As another example, although the wheel turns mechanically, the exact number of officers needed for each event and unplanned events like protests imply that who will be assigned which special event is stochastic but becomes progressively more deterministic as date t approaches.

Thus, at any date τ prior to date t , the portion of $\epsilon_{i,t}$ that is unknown is conditionally independent of connections formed prior to date τ because an officer cannot condition his/her behavior on what is not yet known. The portion of $\epsilon_{i,t}$ that is known may be correlated with connectedness if the officer can manipulate the scheduling process to come into contact with more coworkers and have a better chance at working (or not working) overtime on date t . As a result, if more identifying variation in connectedness comes from the distant past, $Cov(Z_{i,t}, \epsilon_{i,t}|i) = 0$ becomes a more credible assumption.

We compute the proportion of the identifying variation that comes from the recent past by reconstructing connectedness with different amounts of lag. In our baseline specification connectedness is constructed using the 90 days prior to each date t , up to and including $t - 1$. We construct new measures of connectedness that include the 90 days prior to a lagged date $t - 1 - l$, where we set $l = 7, 14, 21$. As connectedness is lagged longer, more of $\epsilon_{i,t}$ becomes unknown to the officer. At the same time, connectedness should become less predictive of overtime, as contact with others farther in the past is less important for obtaining overtime

now. If the coefficient on connectedness becomes statistically or economically small even for short lags, we should be concerned that a large share of identifying variation is contaminated by omitted variable bias. We regress the overtime indicator on lagged connectedness and the controls in our main specification in Appendix Table A.2. We standardized connectedness by dividing by its sample standard deviation, so that the coefficient can be interpreted as the expected increase in overtime probability for a one standard deviation increase in connectedness.

The effect of connectedness lagged by one week, two weeks, three weeks and four weeks is smaller than the original measure by 20.6%, 37.2%, 45.7%, and 56.8% respectively. As we anticipated, lagging connectedness reduces its predictive power. However, the coefficient remains statistically significant at the 0.05 level, and much of the identifying variation remains even when we exclude all potential contact from the last month. Further, the coefficients remain economically significant: even when connectedness is lagged by a month, a one standard deviation increase in connectedness is expected to increase the probability of working overtime by 0.86 percentage points. Relative to a baseline probability of 16.1 percentage points, this is large. This finding confirms the valuations are not completely driven by variation in connectedness from the recent past, but are rather driven by the accumulation of interactions over the entire 90-day period.

Another way to interpret these results is that planned special events¹⁵ are assigned to officers by the wheel at the beginning of the month. Lagging connectedness by one month therefore implies that we are using only connections formed prior to the posting of assignments. Even if an officer could manipulate their work schedule, it would be extremely difficult to know whom to try to work alongside to get a specific shift on a specific date so far in advance. Our results imply that such hard-to-manipulate connections account for 43.2% of the variation in access.

5.3 Estimation

For estimation, we exclude 54 officers who never work overtime. We compute an officer’s base pay rate on each date as the highest rate observed among work-related pay codes that is less than \$34. For dates where an officer does not work, we use the base rate from the most recent work date. We can include both officer and date fixed effects in a nonlinear model because we have a long panel. We employ the analytical bias corrections from Fernández-Val and Weidner 2016 as implemented by Stammann 2017.

¹⁵The government report states in the year prior to our data these represented the vast majority of special events.

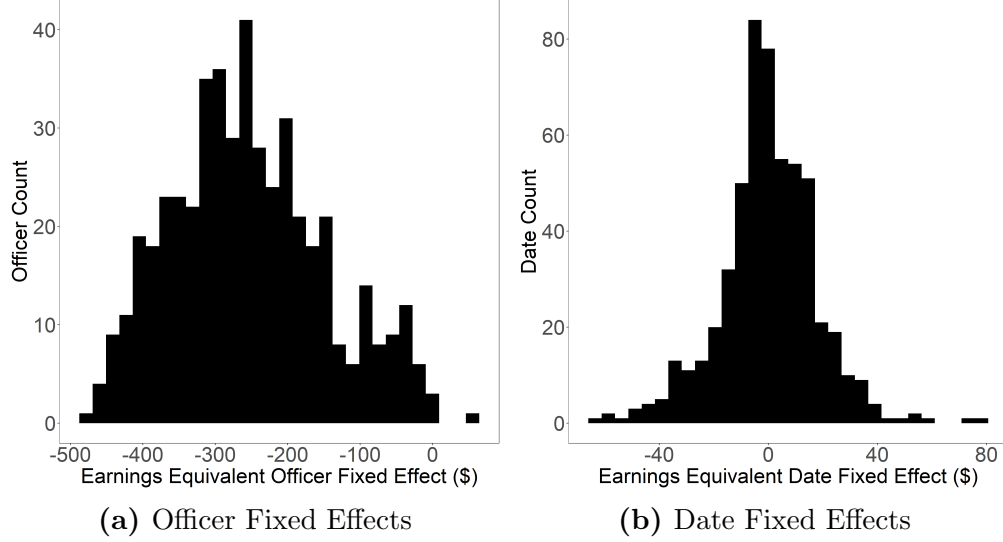
We present coefficients on the time-varying variables in Table 3, along with average partial effects. We also present the estimates from a probit model, to illustrate that the average partial effects implied by our estimates are similar under either parametric assumption. The coefficient on the distance from the wheel median is negative, while the coefficient on potential supplier count is positive, consistent with the existence of access costs generated by the formal wheel system that are mitigated by informal connections. The coefficient on distance from the wheel median is statistically significant at the 0.05 level and economically significant: it implies that the cost of accessing overtime when an officer is maximally distant from the wheel median (low direct access) and has no potential suppliers (low indirect access) is equivalent to \$35.31 in earnings. The coefficient on potential supplier count is also statistically and economically significant: each additional potential supplier reduces access costs by an amount equivalent to \$0.88 in earnings. The average partial effect on seniority rank is small and statistically insignificant. This is reassuring, because it suggests the way we model the wheel and informal connections captures the impacts of seniority rank on overtime access.

Table 3: Model Estimates

Variable	Estimate (Logit)	APE (Logit)	Estimate (Probit)	APE (Probit)
Distance from Wheel	-1.197 (0.130)	-0.105 (0.006)	-0.672 (0.072)	-0.104 (0.006)
Normal Work	0.219 (0.042)	0.019 (0.002)	0.116 (0.023)	0.018 (0.002)
Overtime Wage	0.220 (0.037)	0.019 (0.001)	0.128 (0.019)	0.020 (0.001)
Seniority Rank	-0.015 (0.014)	-0.001 (0.000)	-0.007 (0.008)	-0.001 (0.000)
Supplier Count x Distance from Wheel	0.030 (0.004)	0.003 (0.000)	0.017 (0.002)	0.003 (0.000)

Note: Standard errors are clustered at the officer level. The main specification assumes logistic unobservables and is displayed in the first column.

In Figure 7, we also plot the distributions of officer fixed effects and date fixed effects, denominated in equivalent earnings. The estimates imply that officers value overtime shifts quite differently: in earnings equivalent terms, an officer with a fixed effect at the 75th percentile values an overtime shift \$141.96 more than an officer at the 25th percentile. Heterogeneity in valuations across dates is smaller but still meaningful. If we fix an officer, the date fixed effects imply that a date at the 75th percentile is valued \$19.61 more in earnings equivalent terms than a date at the 25th percentile.

Figure 7: Estimated Fixed-Effect Distributions

Average potential supplier count is positively correlated with preference for overtime, but not perfectly so. We illustrate this in Table 4, which displays the fixed effects (preference estimates) alongside average connectedness and potential supplier count among officers who are likely part of the “cartel.” We define likely members of the cartel as the officers who work more than 300 overtime shifts in the 547 analysis period.

Table 4: The Cartel

Officer ID	OT Shifts		Avg. Connectedness		Avg. Suppliers		Fixed Effect	
	Value	Rank	Value	Rank	Value	Rank	Value	Rank
44	304	179	74.15	20	33.36	104	\$-256.73	229
394	316	180	69.87	48	36.61	27	\$-154.14	86
264	317	181	80.53	5	39.91	4	\$-209.7	154
471	317	181	80.59	4	39.49	7	\$-117.26	59
308	325	182	82.02	3	43.22	2	\$-177.43	112
124	330	183	78.73	6	39.7	5	\$-261.61	239
373	368	184	76.53	9	40.43	3	\$-153.77	84
200	390	185	78.66	7	38.77	10	\$-3.11	3
432	473	186	84.06	2	34.91	53	\$-99	53
537	524	187	89.3	1	43.6	1	\$48.67	1

Note: Overtime valuations, connectedness, and number of overtime shifts worked among likely members of the cartel.

As the government report suggests, the officers who work the most overtime are by and large the most well-connected in the organization. Being well connected usually but not always implies better access to overtime in terms of more potential suppliers. Discrepancies between the measures reflect the fact that access depends not just on how many people

an officer knows but the relative distribution of these people across the wheel of seniority. We finally observe that, with the exception of officer 200 and officer 537, the other 8 likely members of the cartel do not value overtime that much relative to their peers. For example, officer 308 is the third most connected and has the second most potential suppliers and as a consequence works the 6th most overtime shifts. However, officer 308 has a lower valuation of overtime than 111 other officers. Such patterns hint at the gain from alternative assignment mechanisms like shift auctions. By assigning work based on who values it most (or, who dislikes it the least) rather than who has the best access, the organization may increase aggregate officer surplus.

In contrast to the cartel, some officers are estimated to have a strong preference for overtime because they worked a decent amount of overtime despite being poorly connected. For example, there is an officer who has fewer potential suppliers than 478 other officers but who is observed working 131 overtime shifts. As a consequence, this officer is estimated to have the second-highest overtime valuation. If informal trading via the internal network is replaced by a centralized system, this officer would likely work significantly more overtime.

5.4 Validation

We validate our estimates using two pieces of data that were not incorporated in estimation. We show that officer fixed effects, which we interpret as person-specific overtime valuations, are strongly negatively correlated with officer age. We then show that date fixed effects, which we interpret as the average amenity value of working the special events on date t , are highest on dates with a large celebrity presence.

We begin with the officer fixed effects. We plot all officer fixed effects against officer age on January 1, 2015 in Figure A.7. There is a negative correlation between the fixed effects and officer age of -0.51 . On average, an officer a year older values the same overtime shift \$5.97 less, consistent with a labor supply elasticity that is decreasing in age. Similarly, officers observed on family leave during our analysis period have smaller fixed effects on average ($-\$259$ vs $-\$221$). We plot the distribution of fixed effects in Appendix Figure A.8. Thus our estimates are consistent with officers with family concerns having a higher opportunity cost of additional work.

We next consider our estimates of the date fixed effects $\hat{\gamma}_t$, converted to be in terms of earnings equivalents. If we are correctly capturing officer valuations for shifts with our empirical specification, the date fixed effects should reflect the average amenity value of working overtime on a specific date. We residualize the fixed effects so that they are not

mechanically higher on dates when more officers are needed.¹⁶ Thus, the fixed effects can be interpreted as indicating the amenity value of a shift relative to other dates with the same number of officers working overtime.

We then take the 10 dates with the highest adjusted fixed effects, and we read the special event permits listed for each date. We record the most notable event among the list of special events. If our model is capturing officer valuations, the top 10 dates should be those with high amenity value. Appendix Table A.3 shows this is indeed the case.

Of the several hundred dates in the data, the dates with the highest fixed effects had particularly notable special events. Popular press articles indicate that many of these specific events involved large numbers of celebrities or professional athletes. Because we did not use the special-event permit information during estimation, this gives us confidence that our estimates are capturing underlying officer preferences. We emphasize that it is difficult to identify all notable special events, as descriptions are sometimes vague and not all special events are documented by the popular press. For example, many of the special events listed on January 1, 2015 are New Year’s Eve parties with ambiguous descriptions. But parties large enough to require street closures and special-event permits likely also involved celebrities.

We provide additional evidence by examining average adjusted fixed effects for different days of the week and on days when it did or did not rain. Recall that officers have a normal work schedule that they must accommodate when taking on special events. The wheel is not designed around it, and if the special event conflicts with an officer’s normal work, they must take time off using sick or vacation banks to attend. As a result, if our fixed effects reflect average valuations, they should be lower on week days when a larger share of officers need to work their normal shift. Rain is not common in Los Angeles, it can be predicted beforehand, and traffic officers perform most of their duties outdoors. As a result, if the date fixed effects reflect amenity value, they should be lower on dates with rain. We find in Appendix Table A.4 that both patterns hold.

6 Decomposing Overtime Inequality

We use our estimates to decompose the sources of overtime inequality, and show that preference heterogeneity rather than differences in informal networks are the main driver of observed inequality. Wage differences across officers, due to the union-negotiated tenure-based pay scale, tend to reduce overtime inequality.

¹⁶We regress the fixed effects on fixed effects for the number of officers working overtime that day and take the residual.

If we examine our specification for the utility from working an overtime shift, three channels drive differences in the amount of overtime officers work:

$$\underbrace{\rho Wage_{i,t}}_{\text{Pay Scale}} + A_{i,t}(\beta_2 + \underbrace{\beta_3 Z_{i,t}}_{\text{Informal Network}}) + X'_{i,t}\beta_1 + \gamma_t + \underbrace{\alpha_i}_{\text{Non-wage Preferences}} + \epsilon_{i,t}$$

1. **Informal Network.** Well-connected officers will have more potential suppliers and can therefore access more overtime via informal trading.
2. **Pay Scale.** Officers at a higher pay scale will have a higher base rate, which will increase the pay they receive from an overtime shift.
3. **Non-Wage Preferences.** Officers receive different amenity value from working overtime, reflecting both canonical differences in leisure preferences and differences in preferences for celebrity contact.

To quantify the contribution of each channel, we shut down combinations of channels and compute the expected number of overtime shifts worked by each officer. We then compute the share of overtime worked by the top 10% of officers and compare it to the observed share. When shutting down each channel, we keep the expected number of overtime shifts worked constant by adding the appropriate constant to each officer-date's utility. If $u_{i,t}$ is the deterministic part of utility after the appropriate channels are shut down, the appropriate constant, call it c^* , solves the following equation:

$$\sum_{i=1}^N \sum_{t=1}^T \Lambda(u_{i,t} + c^*) = \sum_{i=1}^N \sum_{t=1}^T W_{i,t},$$

where $\Lambda(\cdot)$ is the standard logistic CDF.

Because the expected number of shifts worked by each officer is non-linear in the various channels, the order in which channels are shut down is important. For this reason we present three measures. First, we compute the change in inequality from just shutting down the focal channel but keeping the other two channels active. Second, we compute the change in inequality from shutting down all channels except the focal channel. Finally, we compute a Shapley decomposition meant to capture the average impact of shutting down the focal channel across all possible initial sets of active channels. Specifically, we equally weight the change from shutting down the focal channel from all four of the possible initially active channels: all channels active, focal channel and non-focal channel 1 active, focal channel and non-focal channel 2 active, only the focal channel active. We present the results in Table 5.

Several patterns emerge. First, the informal network is responsible for only a small

amount of inequality. Second, non-wage preferences are the main driver of observed inequality, and removing preference differences but keeping all other differences intact would reduce the share of overtime worked among the top 10% of officers by 7.2 percentage points, which is large relative to the observed share of 36.3%. Third, wage differences that come through differences in tenure tend to reduce inequality: equalizing officer wages would increase overtime inequality.

The decomposition exercise reveals that eliminating access differences due to informal networks is unlikely to noticeably improve overtime inequality. Indeed, the exercise suggests that it may increase inequality. Table 5 shows that when non-wage preferences are left unchanged but the other two channels are shut down, the share of overtime worked by the top 10% of officers rises by 4.3 percentage points.

Table 5: Decomposing Overtime Inequality

Channel	Remove All Others	Remove Just Channel	Shapley Average
Informal Network	0.063	-0.006	-0.001
Non-Wage Preferences	0.043	0.072	0.038
Wages	0.081	0.042	0.023

7 Counterfactuals

This section compares the status quo of informal trading to two counterfactual shift auctions and random assignment. We show that informal trade achieves most of the allocative efficiency gains that would occur from moving from random assignment to a centralized auction which maximizes allocative efficiency. We also show that informal trade generates similar inequality and worker surplus. Under all assignment mechanisms, we hold fixed the number of overtime opportunities on each date.

7.1 Overtime Assignment Mechanisms

We consider four assignment mechanisms. In all counterfactual simulations, we do not incorporate dynamic worker behavior. Specifically, we do not adjust potential supplier count based on who works in each simulation, and our preference specification does not incorporate dynamic preferences for overtime.

7.1.1 Informal Trading

The status quo is an informal trading system. As this is the system under which the observed data are generated, we discuss it extensively in Section 2. We use our estimated model to simulate informal trading. We redraw the idiosyncratic utility shocks $\epsilon_{i,t}$ from a standard logistic distribution 2,500 times. For each draw we assume the k officers with the highest utility (including the network and access components) work the k overtime slots. Note that because informal trade is simulated, the average level of inequality will not exactly match the level of inequality observed in the data.

7.1.2 Uniform-Wage Shift Auctions

The first alternative overtime assignment mechanism we consider is a uniform-wage shift auction. On each date t , the LADOT auctions k_t overtime slots, where k_t is the number of officers working overtime in the data on date t . On each date, all active officers simultaneously submit one sealed wage bid.¹⁷ The officers that submit the lowest k_t wage bids are assigned the shift, and they are paid the $k_t + 1$ lowest wage, i.e. the first wage that did not win. This is a uniform-wage shift auction because all officers who work on date t are paid the same overtime wage. We assume that since the informal secondary market is replaced by an auction, the access cost of obtaining a shift is 0. We implement this in our specification by setting $A_{i,t} = 0$ for all officers on all dates. Thus it is as if all officers are being directly given a shift by the wheel if they bid a low enough wage. Because $A_{i,t}$ multiplies potential suppliers in our specification of access costs, this also shuts down all variation in access costs across officers.

Uniform-wage shift auctions have the property, which we prove in Appendix Section C, that it is a weakly dominant strategy for officers to bid truthfully. That is, to bid a wage that is the negative of their valuation of working the shift. The shift is then allocated to the k officers with the least dis-utility from work. We run this auction for each date in the sample among all active officers drawing $\epsilon_{i,t}$ from a standard logistic distribution. We set k , the number of shifts to be assigned each date, to be exactly the number of overtime shifts observed in the data on that day. We simulate the auction 2,500 times.

¹⁷Although this is a multi-unit auction we assume that each officer can work at most one overtime shift per date.

7.1.3 Uniform-Markdown Shift Auctions

Uniform-markdown shift auctions are identical to uniform-wage shift auctions, except that officers bid a markdown from their status quo overtime rate.¹⁸ The officers who submit the lowest k_t wage markdown bids are assigned the shift, and they are paid their status quo overtime rate less the $k_t + 1$ largest wage markdown, i.e. the first markdown bid that did not win. This is a uniform-markdown shift auction because all officers who work on date t are paid the same markdown from their status quo wage but are not paid the same wage. Interesting properties of uniform-markdown auctions are discussed in Appendix Section D. We simulate uniform-markdown auctions in the same way as uniform-wage auctions.

7.1.4 Random Assignment

The final alternative we consider is random assignment. Under the random assignment benchmark, we fix the number of shifts that need to be filled on each date. We then randomly order the officer active on that date, assign the shift to those officers, and assume they worked the shift.

7.2 Allocative Efficiency and Overtime Inequality

We now study how each system impacts overtime inequality and allocative efficiency. We measure allocative efficiency as the total non-wage utility of those who work overtime under an allocation. That is for an allocation $W = \{W_{i,t}\}_{i=1}^{N,T}$ allocative efficiency is

$$E(W) := \sum_{i,t} W_{i,t} \cdot (X'_{i,t} \beta_2 + \gamma_t + \alpha_i + \epsilon_{i,t}).$$

Our measure intentionally excludes transfers (wages) and access costs incurred during informal trade to focus only on the non-wage utility obtained by workers from each overtime opportunity. We first highlight that because the uniform-wage auction always assigns overtime to the officers with the highest non-wage utility from overtime, it achieves the maximum allocative efficiency for any fixed number of overtime shifts. We measure inequality by computing the share of overtime shifts assigned to the top 10% of officers. A more equitable system distributes a smaller share of overtime to the top 10% of officers. By construction, random assignment minimizes inequality. Because shifts are not divisible and we require that officers be active on the date they are assigned, random assignment will not achieve perfect equality.

¹⁸The status quo rate is 1.5 times their base rate, which comes from a union-negotiated pay scale.

We report the results in Table 6. We aggregate the results from all simulations by reporting the mean, 5th percentile and 95th percentile. When comparing systems, we focus primarily on the mean of a performance metric. The effective lower bound on the share of overtime worked by the top 10% of officers is 0.123, achieved by random assignment. The effective upper bound on allocative efficiency is \$ $-9,187,675$ in earnings equivalent terms, achieved by the uniform-wage auction. There is an equity-efficiency trade-off: the uniform-wage auction achieves the highest allocative efficiency and generates the most inequality, while random assignment achieves the least inequality but generates the lowest allocative efficiency.

Table 6: Inequality and Allocative Efficiency under Different Mechanisms

Assignment Mechanism	Allocative Efficiency			Share OT by Top 10		
	p5	Mean	p95	p5	Mean	p95
Uniform-Markdown Auction	\$ -9,495,984	\$ -9,480,403	\$ -9,464,744	0.331	0.334	0.337
Uniform-Wage Auction	\$ -9,202,578	\$ -9,187,675	\$ -9,173,061	0.370	0.372	0.375
Informal Trading (Status Quo)	\$ -9,500,206	\$ -9,482,592	\$ -9,469,604	0.335	0.338	0.341
Random Assignment	\$-13,829,940	\$-13,805,425	\$-13,780,589	0.122	0.123	0.124

Relative to random assignment, informal trade increases allocative efficiency by \$4.32 million in earnings equivalent terms. This represents 93.6% of the gap between random assignment and the maximum achieved by the uniform-wage auction. Informal trade is also more equitable than the uniform-wage auction, resulting in 33.8% of overtime shifts assigned to the top 10% of officers compared to 37.2% under the uniform-wage auction. Informal trade achieves almost the same allocative efficiency and inequality as the uniform-markdown auction. In terms of both equity and efficiency, informal trade performs relatively well, even when bench-marked against a centralized and perfectly implemented auction.

7.3 Wages and Worker Surplus

Although informal trading, uniform-wage and uniform-markdown auctions achieve similar allocative efficiency, how they impact workers and the LADOT also depends on wages. For this reason, we consider how each system impacts the total wage bill and worker surplus. For each system, we report the 5th percentile, the mean, and the 95th percentile across all simulations in Table 7. In discussions, we focus on the mean.

Three patterns are worth noting. First, the uniform-wage auction results in the lowest wage costs and the greatest worker surplus. This is not surprising, as the uniform-wage auction assigns shifts to the officers that value them most and extracts the surplus generated via lower wages. Second, the changes in wage costs and worker surplus from switching

Table 7: Wage Costs and Worker Surplus under Different Mechanisms

Assignment Mechanism	Overtime Wage Costs			Worker Surplus		
	p5	Mean	p95	p5	Mean	p95
Uniform-Markdown Auction	\$10,955,551	\$10,969,000	\$10,982,951	\$ 1,475,278	\$ 1,486,622	\$ 1,497,828
Uniform-Wage Auction	\$10,841,979	\$10,855,102	\$10,868,561	\$ 1,647,023	\$ 1,659,275	\$ 1,671,983
Informal Trading (Status Quo)	\$11,086,336	\$11,092,442	\$11,098,563	\$ 1,472,176	\$ 1,486,658	\$ 1,499,861
Random Assignment	\$11,118,992	\$11,125,869	\$11,132,777	\$-2,704,416	\$-2,680,961	\$-2,657,138

from informal trade to the uniform-wage auction are modest. Third, the uniform-markdown auction incurs higher wage costs and delivers less worker surplus than the uniform-wage auction because it advantages officers higher up on the pay scale, which requires paying higher wages and also distorts allocations.

We take a moment to specifically compare the uniform-wage auction and informal trade. The wage cost reduction from moving from the status quo to uniform-wage auctions is \$237,340 or 2.14%. In dollar terms this is quite similar to the corresponding improvement in allocative efficiency (\$294,917). The increase in worker surplus is \$172,617 or 11.6%. This is equivalent to paying each officer \$0.70 more per hour for overtime, relative to the hours-weighted average overtime wage of \$40.38 that is paid out in the data.

We emphasize that although aggregate outcomes under some pairs of systems are similar, outcomes for individual officers can vary quite substantially across systems. We illustrate this by computing wages, overtime shifts and worker surplus for the likely cartel members in Appendix Section E.

8 Understanding the Efficiency of Informal Trading

Our results indicate that informal and frictional trade achieves similar allocative efficiency as formal and frictionless auctions. This section performs an additional empirical exercise to understand why. We find that informal trade performs well because preference heterogeneity dominates access costs by an order of magnitude, and further, even if access costs were scaled up, the endogeneity of the trading network causes a version of the Coase Theorem to hold.

The exercise we perform starts from our estimating equation:

$$W_{it} = \begin{cases} 1 & \text{if } \rho Wage_{i,t} + X'_{i,t}\beta_1 + \gamma_t + \alpha_i + \epsilon_{i,t} + A_{i,t}(\beta_2 + \beta_3 Z_{i,t}) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The coefficient on the angular distance from our proxy of the wheel's position (β_2) represents access costs. The coefficient on potential supplier count interacted with the angular distance

(β_3) represents the access cost reductions that come from an officer’s informal network. Empirically, we find the dominant determinant of officer preferences are the officer fixed-effects α_i . Thus, $Cov(\alpha_i, Z_{i,t})$ represents the allocative role over the informal network: how much are preferences correlated with access cost reductions? The efficiency of allocations is determined by all three channels.

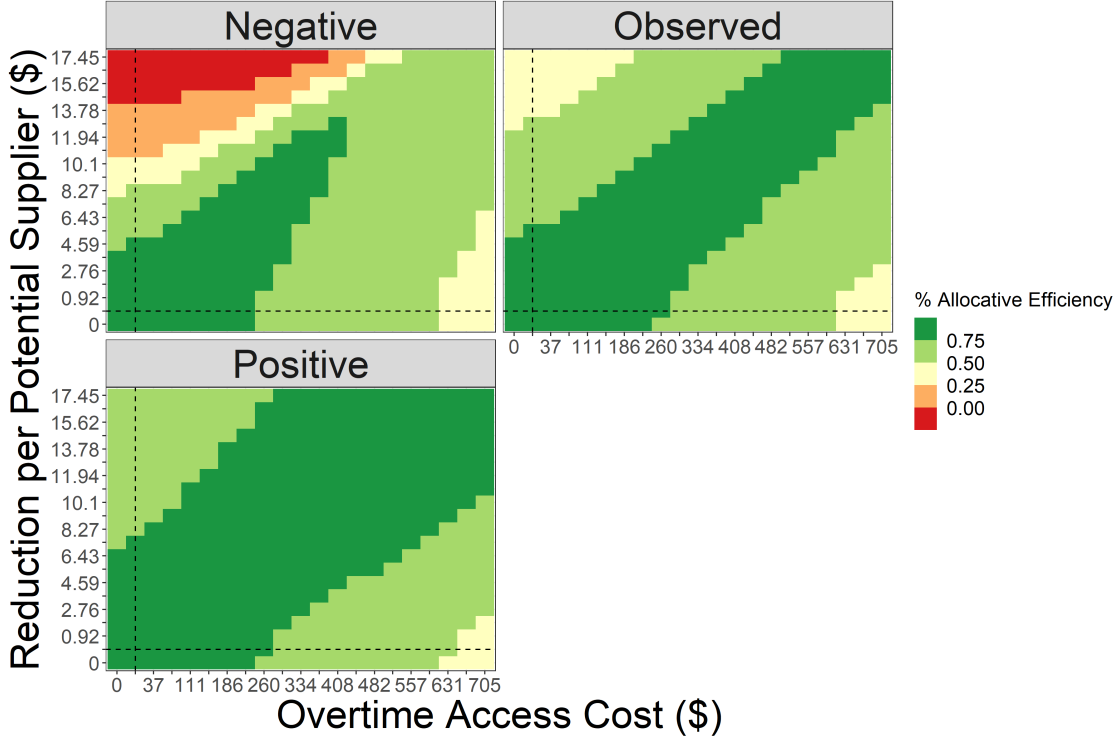
To assess the quantitative role of each channel, we perform counterfactual simulations where we vary β_2 , β_3 and $Cov(\alpha_i, Z_{i,t})$ above and below their estimated values. We vary β_2 from 0 (no access costs) to -23.5 (20 times larger). We vary β_3 from 0 (no informal network access cost reductions) to 0.593 (20 times larger). We consider three regimes for $Cov(\alpha_i, Z_{i,t})$. In the *positive regime*, we reassign the highest number of potential suppliers on each date to the officer with the highest fixed-effect. In the *observed* regime, we keep the data as is. In the *negative* regime, we reassign the highest number of potential suppliers on each date to the officers with the lowest fixed-effect. We perform every combination of access cost magnitude and regime, and compute the average allocative efficiency over 2,500 simulations.

The results are depicted in Figure 8, where allocative efficiency is measured as a percentage of the gap in allocative efficiency between random assignment and the maximum achieved by a uniform-wage shift auction. We point where the two dotted liners intersect is the efficiency of informal trade under the estimated parameters for reference. Efficiency values are deliberately binned into coarse color categories for visual comparison of regimes. For a more fine-grained version, see Appendix Figure A.9.

Holding regime and the reduction per potential supplier fixed, efficiency remains similar as overtime access costs (β_2) increase. It is necessary to more than triple access costs before efficiency begins noticeably declining. In the same way, if regime and access costs are held fixed, it is necessary to increase the reduction per potential supplier (β_3) by several times before efficiency is impacted. Thus, informal trade is efficient partly because access costs and the mitigation that occurs through the network is small relative to preference differences (α_i).

Another notable pattern is that there is a region of green efficient outcomes along the approximate 45 degree line as β_2 and β_3 scale together. This region is largest in the positive regime, reflecting the fact that we artificially tied together valuations and the number of potential suppliers in that regime. The region is smallest and disappears most rapidly for the negative regime, because preference for work is not correlated with a privileged position in the network of trade. The observed data is more qualitatively similar to the the positive regime than the negative regime. Thus, the observed network appears to encode the preferences of officers. If access costs were counterfactually increased and the mitigation from

Figure 8: Mechanisms Behind the Efficiency of Informal Trade



the network was proportionally increased (so that the network played a more important role in allocation), informal trade would continue to be relatively efficient because of the positive correlation between potential supplier count and preference for additional work.

We interpret these results as an example of an endogenously formed network creating the conditions for a version of the Coase Theorem to hold. An officer gains a potential supplier by working alongside someone they do not typically work alongside, and this happens primarily by working a special event. This creates a virtuous cycle: working more special events creates more potential trading partners which makes working special events easier in the future. Thus, the encoded network lowers transaction costs for exactly the right people, leading to efficient allocations.

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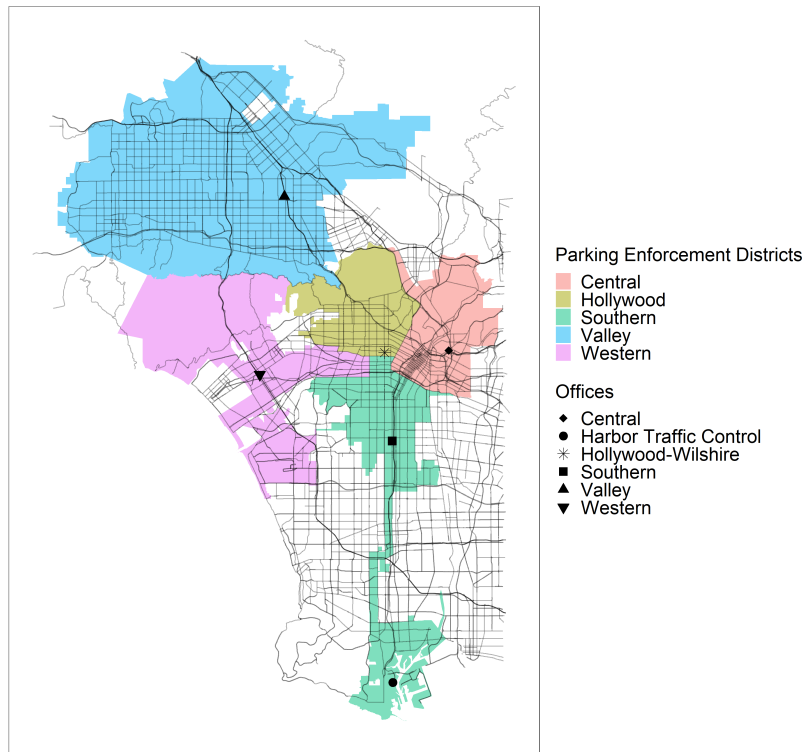
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Appendix

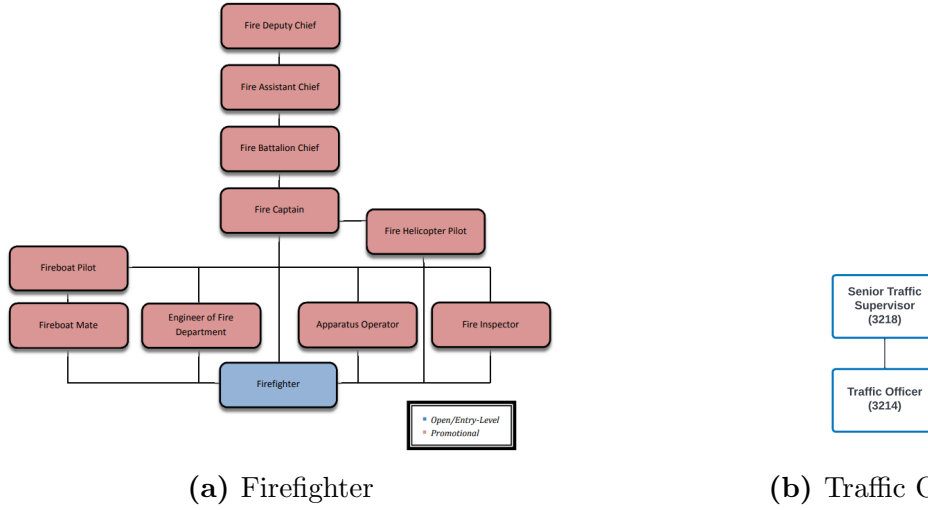
A Additional Tables and Figures

Figure A.1: Parking Enforcement Districts and Offices



Note: Map created using Shapefiles from City of Los Angeles [2023](#) and office addresses obtained from Los Angeles Department of Transportation [2025](#).

Figure A.2: Career Progressions for Firefighters and Traffic Officers

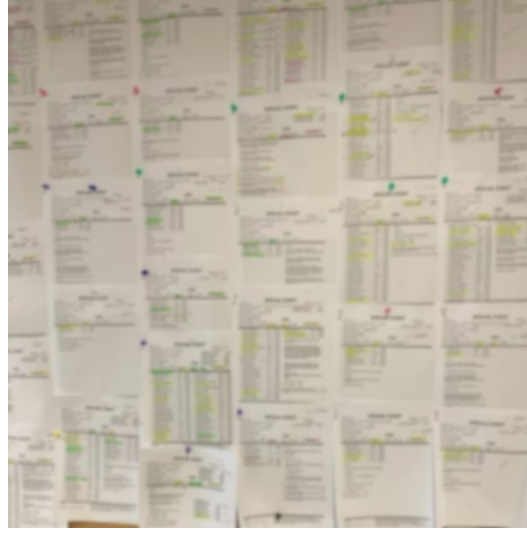


Note. Los Angeles provides a career ladder describing potential promotions and lateral movements for each employee classification (City of Los Angeles Personnel Department 2025). Panel A shows the ladder for firefighters, while Panel B shows the equivalent ladder for traffic officers.

Table A.1: Notable Special Events

Event Name	Date
57th Annual Grammy Awards	2/8/2015
Faith Evans' Grammy After Party	2/8/2015
The 87th Academy Awards (Oscars)	2/22/2015
Katy Perry Prismatic World Tour	3/26/2015
Ice Cube, MC Ren and Yella Concert (NWA Reunion)	6/29/2015
Taylor Swift Concert (1989 World Tour)	8/21/2015
Billionaire Dennis Tito's Holiday Party	12/11/2015

Figure A.3: Implementation of the Wheel System

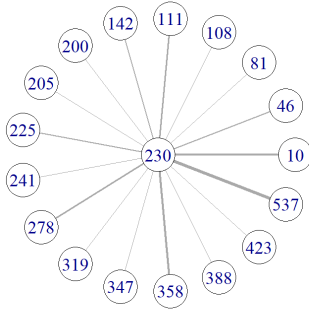


(a) Bulletin Board

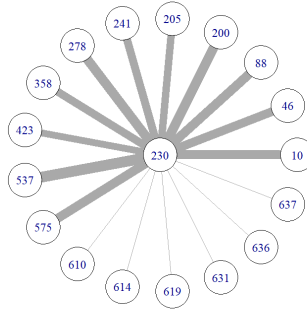
(b) Wheel Assignment

Note. The overtime assignments are considered public record by the City of Los Angeles. Photographs are provided by an internal source within the LADOT.

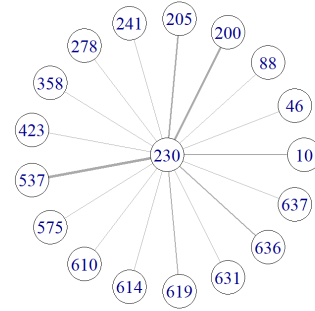
Figure A.4: Variation in Connectedness Across Time



(a) January 1, 2015



(b) July 1, 2015



(c) January 1, 2016

Note. The panels show the connections of Officer 230 at three different points in time. The thickness of each edge represents the strength of the connection measured by potential contact.

Figure A.5: Distribution of Potential Supplier Count

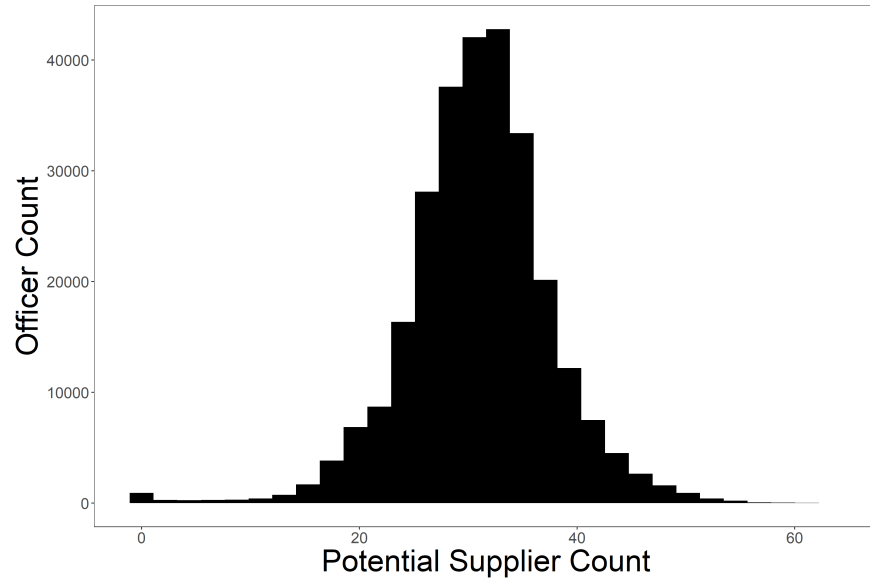
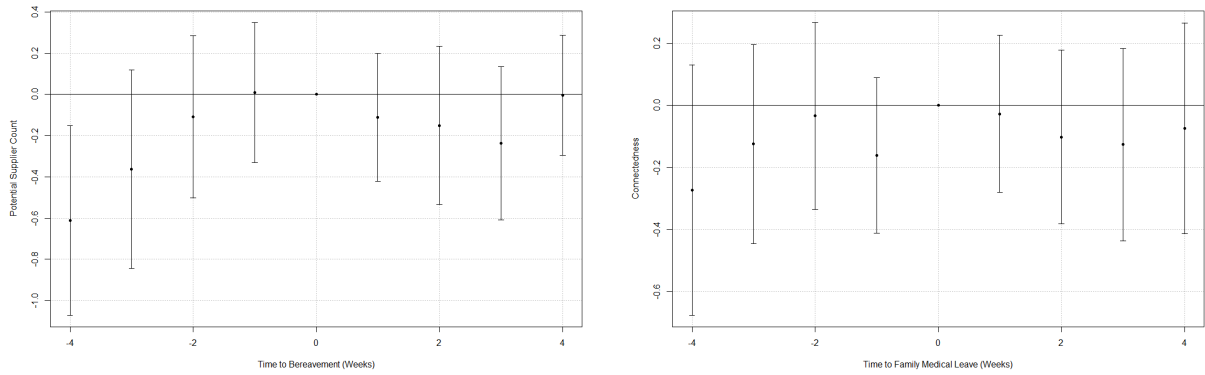


Figure A.6: Own Personnel Changes and Potential Supplier Count



(a) Own Bereavement

(b) Own FMLA

Table A.2: The Role of Lagged Connectedness on the Extensive Margin of Overtime

Dependent Variable: Model:	(1)	(2)	Overtime (3)	(4)	(5)
<i>Variables</i>					
Std. Connectedness	0.0199*** (0.0043)				
Std. Connectedness (Lag 7)		0.0158*** (0.0041)			
Std. Connectedness (Lag 14)			0.0125** (0.0040)		
Std. Connectedness (Lag 21)				0.0108** (0.0039)	
Std. Connectedness (Lag 28)					0.0086* (0.0038)
Distance from Wheel Median	-0.0258*** (0.0040)	-0.0259*** (0.0040)	-0.0265*** (0.0040)	-0.0271*** (0.0041)	-0.0274*** (0.0041)
Normal Work	0.0204*** (0.0038)	0.0196*** (0.0038)	0.0198*** (0.0038)	0.0201*** (0.0038)	0.0202*** (0.0039)
Seniority Rank	-0.0027* (0.0013)	-0.0029* (0.0014)	-0.0030* (0.0014)	-0.0031* (0.0014)	-0.0032* (0.0014)
ot_rate	0.0089*** (0.0018)	0.0088*** (0.0018)	0.0088*** (0.0019)	0.0086*** (0.0019)	0.0085*** (0.0019)
<i>Fixed-effects</i>					
Officer	Yes	Yes	Yes	Yes	Yes
Date	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
R ²	0.24791	0.24876	0.24853	0.24896	0.24934

Clustered (Officer) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05*

Note: Standard errors are clustered at the officer level.

Figure A.7: The Association Between Officer Fixed Effects and Age

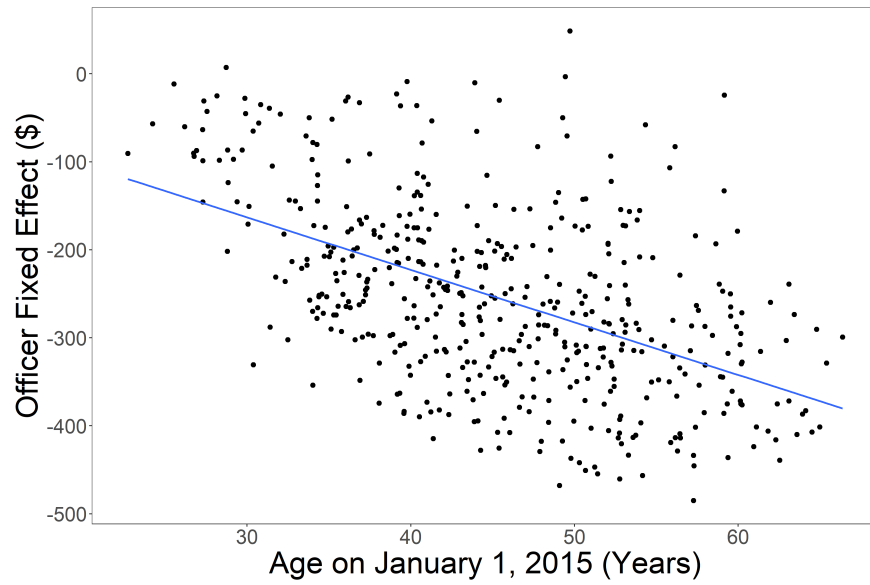


Figure A.8: Officer Fixed Effects and Family Leave

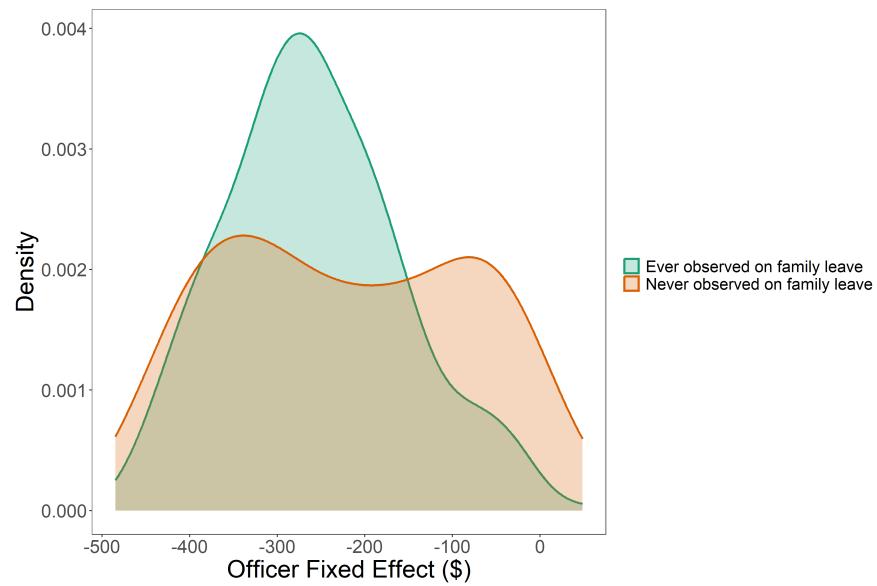


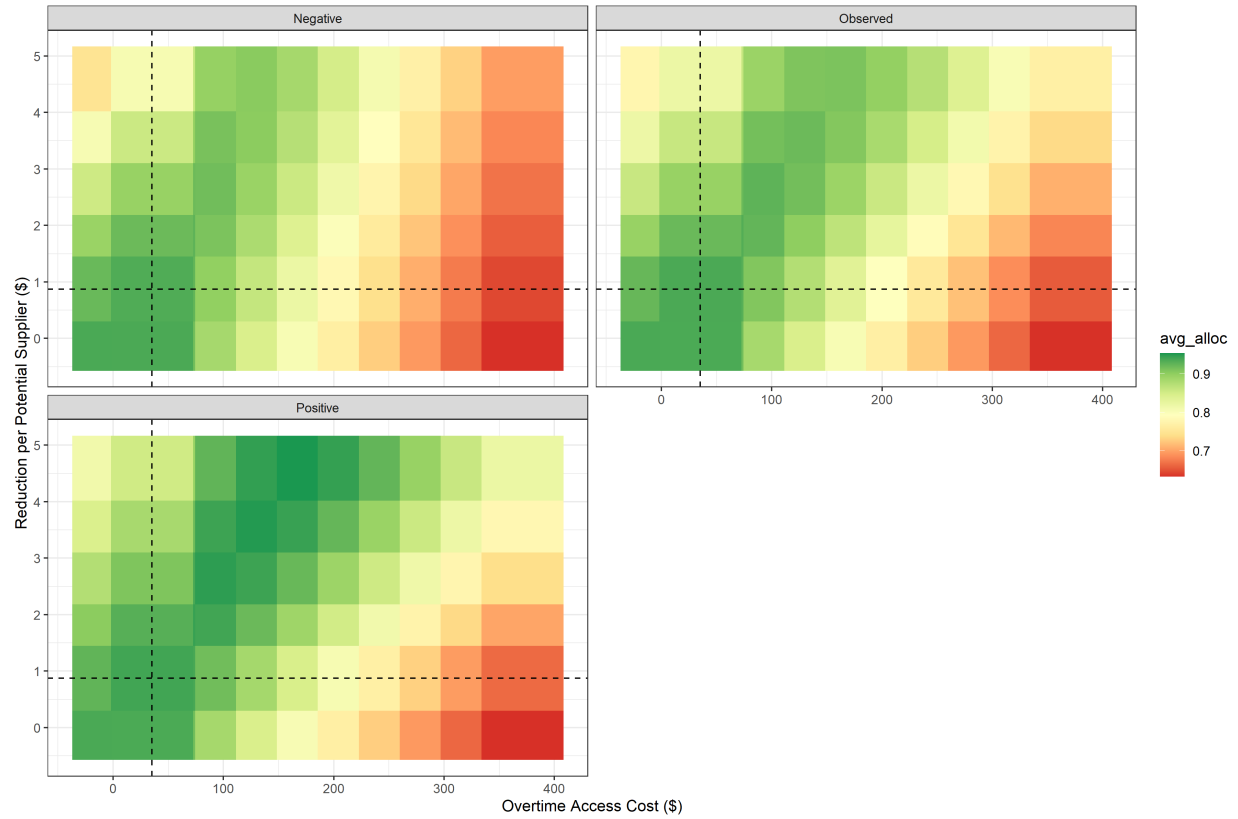
Table A.3: Highest Adjusted Fixed Effects and Notable Special Events

Date	Notable Event	Fixed Effect Earnings Equivalent	Total Overtime
February 22, 2015	2015 Academy Awards (Oscars)	\$12.82	84
April 11, 2015	2015 MTV Movie Awards	\$11.90	101
January 01, 2015	New Year's Eve Parties	\$11.32	65
February 21, 2015	Oscar's Night Before Party w/ George Clooney and Reese Witherspoon	\$10.99	105
January 03, 2015	New Year's Race	\$10.95	75
April 26, 2015	2015 Daytime Emmy Awards	\$10.88	77
April 19, 2015	LACMA 50th w/ Will Ferrell and Jim Carrey	\$10.80	95
February 16, 2015		\$10.71	72
January 19, 2015		\$10.27	135
January 15, 2015		\$9.74	67

Table A.4: Average Adjusted Fixed Effect by Day of Week and Rain

Type of Day	Number of Dates	Avg. Fixed Effect	Avg. Overtime Shifts
No Rain	502	\$0.02	70.27
Rain	45	\$-0.19	63.29
Thursday	79	\$-0.26	69.75
Friday	78	\$-0.41	75.42
Saturday	78	\$1.83	71.78
Sunday	78	\$1.77	67.05
Monday	78	\$-0.45	68.29
Tuesday	78	\$-1.21	67.49
Wednesday	78	\$-1.27	68.08

Figure A.9: Mechanisms Behind the Efficiency of Informal Trade: Continuous Version



Note. This figure is an additional version of Figure 8 where the heatmap is allowed to vary more continuously. This shows heterogeneity across the parameter values but makes it more difficult to see the qualitative similarities between the observed and the positive regimes.

Figure A.10: Peer Impacts on Connectedness: Two-Way Fixed Effects

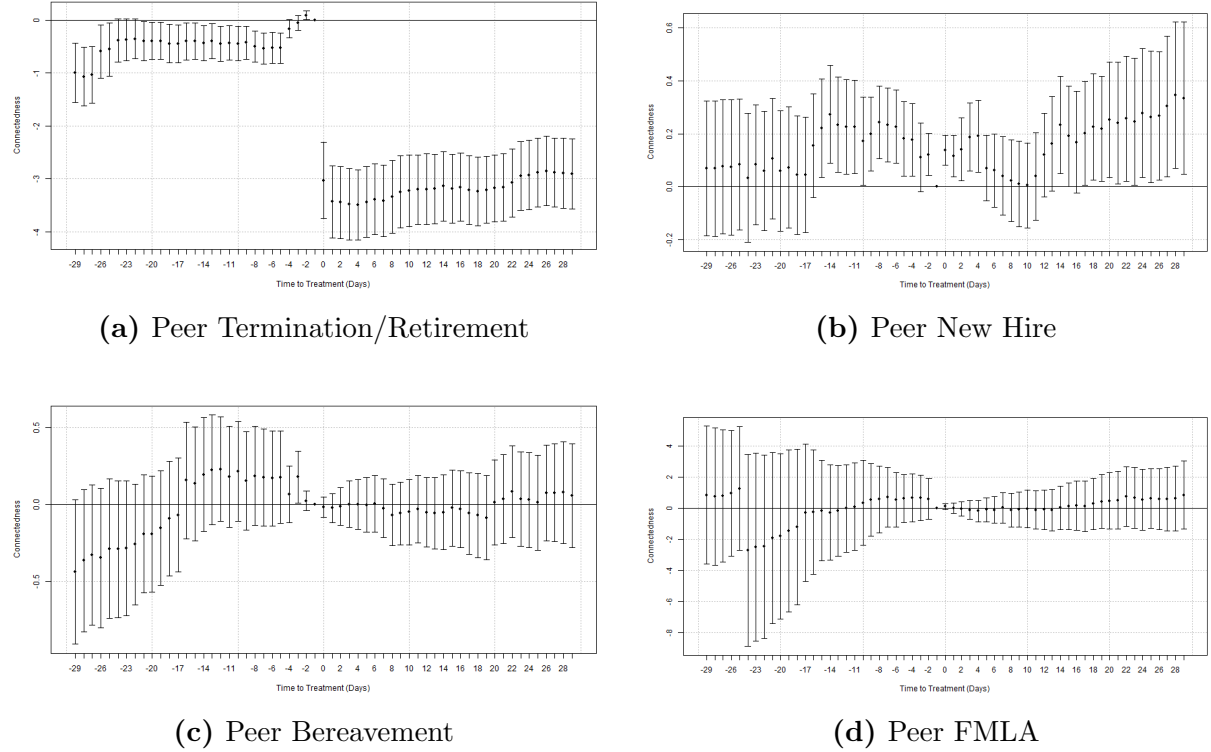
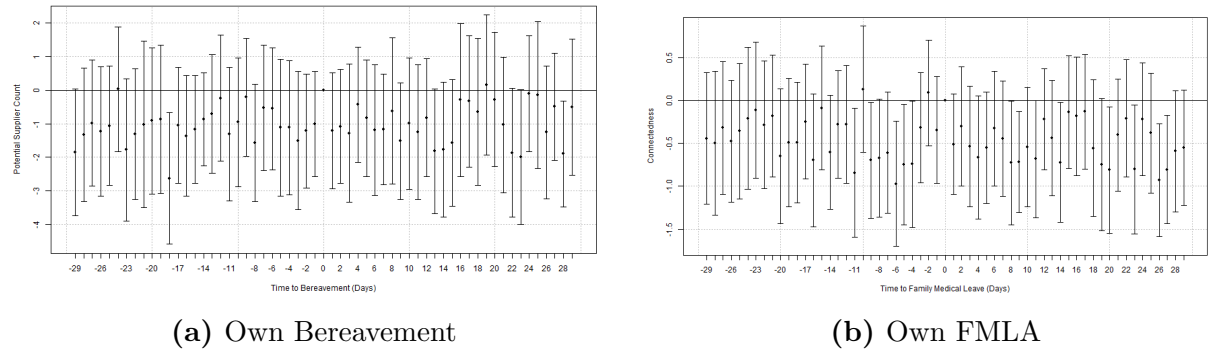


Figure A.11: Own Personnel Changes on Potential Supplier Count: Two-Way Fixed Effects



B Peer Personnel Changes and Connectedness

In this section, we show how personnel changes of surrounding officers translate into the connectedness of a focal officer. The demonstration and associated tables and figures are not tests of identification. Rather, they show how variation in our network centrality measure is driven by real changes in the organization. We show that an officer's potential supplier

count falls when a coworker terminates/retires and rises when an officer is hired in their division. We use connectedness rather than potential supplier count because connectedness is not contaminated by the high-frequency noise induced by the wheel’s sometimes daily rotations.

We create a treatment indicator where an officer is considered treated if they worked in the same division on the same day as the terminated or retired officer at least one of the 90 days prior to termination or retirement. We create another treatment indicator where an officer is considered treated if they worked in the same division as the new hire on the new hire’s first day. The outcome is lagged connectedness. We estimate the average treatment effect on the treated using the Sun and Abraham 2021 estimator as well as two-way fixed effects and report the results in Table B.1. Note that the Sun and Abraham 2021 estimator required aggregating the data to be at the weekly level due to numerical and sample size problems.

Table B.1: Peer Impacts on Connectedness

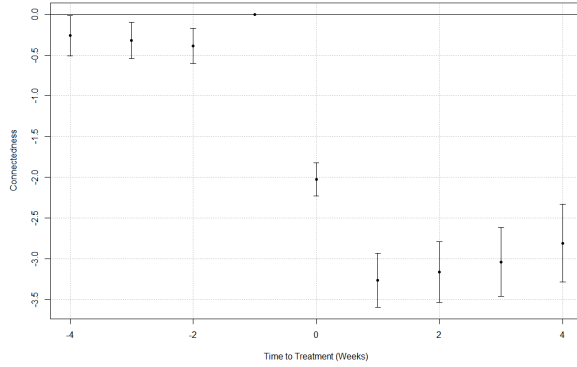
Specification	Estimator	ATT	Std. Error	p-value	Observations
Termination	TWFE	-1.872	0.328	< 0.001	268,186
	Sun Abraham	-2.734	0.411	< 0.001	38,785
New Hire	TWFE	0.141	0.250	0.574	263,571
	Sun Abraham	0.199	0.255	0.435	38,086
FMLA	TWFE	6.099	2.334	0.011	31,907
	Sun Abraham	0.389	0.979	0.692	4,627
Bereavement	TWFE	1.080	0.416	0.010	239,323
	Sun Abraham	-0.466	0.347	0.180	34,618

Note: Standard errors are clustered at the officer level.

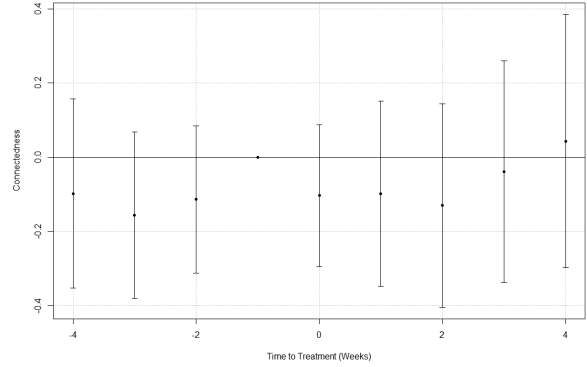
To further interpret these results, we provide event study plots for the Sun and Abraham 2021 estimates in Figure B.1.¹⁹ Taken together, the estimates and plots show that the termination of a peer leads to a large and statistically significant reduction in connectedness. Recovery of connectedness occurs gradually over several weeks as the officer is exposed to new peers. There is little evidence that a new hire impacts connectedness. Peer bereavement and peer FMLA do not appear to have an effect when only the ATT is considered, but the event study figures indicate there may be some form of anticipation where connectedness falls prior to a peer taking leave.

¹⁹See Appendix Figure A.10 for the two-way fixed effects event study plots.

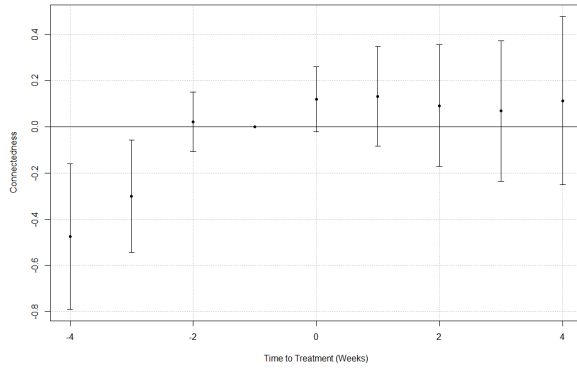
Figure B.1: Peer Personnel Changes and Connectedness



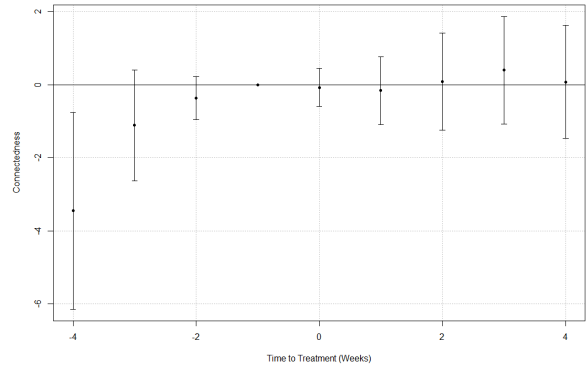
(a) Peer Termination/Retirement



(b) Peer New Hire



(c) Peer Bereavement



(d) Peer FMLA

C Uniform-Wage Shift Auctions: Bidding Strategies

We now solve for the equilibrium bidding strategies of officers. Regardless of what others do, each officer's weakly dominant strategy is to bid the negative of their valuation. To see this, consider an officer with valuation x , denominated in wage equivalents, who bids wage $-x$. Fix the $k + 1$ lowest wage bid at y . First, suppose $-x < y$. In this case, the officer wins the shift and receives wage y obtaining positive utility $y + x > 0$. Any change of bid that is not greater than y has no impact on outcomes and leaves the officer indifferent. Bidding strictly above y causes the officer to no longer win the shift, strictly decreasing utility. Second, suppose $-x \geq y$. In this case, the officer does not win the shift and obtains utility 0. All alternative bids greater than y have no impact on outcomes and leave the officer indifferent. Any bid x' strictly below y causes the officer to win the shift, receive wage y and obtain negative utility because $y + x' < y + x \leq 0$. As a result, officers will bid the negative of their valuation, and the officers with the k highest non-wage valuations for overtime will work

overtime.

D Uniform-Markdown Shift Auctions

Two important properties of uniform-markdown auctions are discussed in this section.

Truthful Bidding. In a uniform-markdown shift auction, officers continue to bid their true value for the shift as under a uniform-wage shift auction. We provide a proof at the end of this section, but the argument is very similar to the one provided for uniform-wage shift auctions. The key strategic difference between uniform-wage and uniform-markdown auctions is that officer valuations for the same overtime shift are different under each format, meaning that who wins and works the shift will be different under each system.

Preservation of Pay Scale. A feature of uniform-markdown shift auctions is that they preserve the status-quo differences in wages across officers. In our setting, these differences are due to the seniority-based pay steps negotiated by the union and outlined in the MOU with the City of Los Angeles. The pay steps are preserved because all the officers who win the auction on a particular date are paid the same markdown from their status quo wage. This feature also implies that, relative to a pure shift auction, the uniform-markdown shift auction implicitly advantages senior officers with a higher pay scale. For this reason, we view the uniform-markdown auction as more feasible than the uniform-wage auction.

Proof of Truthful Bidding. Consider an officer with a non-wage valuation x and a statutory overtime wage of w . Their total valuation, denominated in wage equivalents, of the shift in a uniform-markdown shift auction will be $\tilde{x} := x/\rho + w$.

We wish to prove that it is weakly dominant for the officer to bid a markdown of $\tilde{x}$ even if all the bids of other officers are known. Fix the $k + 1$ highest markdown bid to be y . First, suppose $\tilde{x} > y$. In this case, the officer wins the shift and has their wage marked down by y obtaining positive utility because $\tilde{x} - y > 0$. Any change of bid not smaller than y has no impact on the officer's outcome and leaves the officer indifferent. Bidding a markdown strictly below y causes the officer to no longer win the shift, generating 0 utility and making the officer strictly worse off. Second, suppose $\tilde{x} \leq y$. In this case, if the officer does as we suppose and bids $\tilde{x}$, they do not win the shift and obtain utility 0. All alternative bids less than y have no impact on the officer's outcomes and leave the officer indifferent. Any markdown bid x' strictly above y causes the officer to win the shift and have their wage marked down by at least x' . This generates strictly negative utility because: $\tilde{x} - x' < \tilde{x} - y \leq 0$. Thus, it is a weakly dominant strategy for all officers to bid a markdown equal to their total valuation (which includes their statutory overtime wage).

E Impacts on Likely Cartel Members

Although the aggregate impacts of the counterfactuals are modest in some dimensions, the impacts on individual officers can be quite large. At a high level, uniform-wage shift auctions transfer worker surplus from officers that do not value overtime to officers that do value overtime. We illustrate this by showing the estimated overtime shifts, overtime pay, and worker surplus of the 10 likely cartel members in Table E.1. Switching to uniform-wage shift auctions reduces the worker surplus of the two likely cartel members that have weak preferences for overtime (smaller fixed effects) and increases the worker surplus of the eight likely cartel members that have strong preferences for overtime (larger fixed effects).

Table E.1: Counterfactual Impacts on Likely Cartel Members

Officer ID	FE Rank	Overtime Shifts			Overtime Pay			Worker Surplus		
		Informal	Wage	Markdown	Informal	Wage	Markdown	Informal	Wage	Markdown
537	1	524	540	521	\$126,985	\$149,082	\$124,532	\$55,315	\$73,794	\$53,293
432	53	473	393	467	\$144,825	\$110,203	\$141,593	\$34,616	\$22,273	\$33,494
200	3	390	486	382	\$95,545	\$134,849	\$92,509	\$22,224	\$38,327	\$21,423
373	84	368	249	359	\$113,753	\$70,611	\$109,972	\$19,255	\$10,492	\$18,536
124	239	329	271	311	\$96,451	\$76,557	\$90,201	\$15,801	\$11,832	\$14,510
308	112	325	260	302	\$95,150	\$73,685	\$87,536	\$15,530	\$11,210	\$13,829
264	154	316	263	305	\$92,822	\$74,475	\$88,346	\$14,823	\$11,347	\$14,005
471	59	317	262	303	\$93,029	\$74,110	\$87,877	\$14,866	\$11,266	\$13,924
394	86	317	202	311	\$98,212	\$57,569	\$95,352	\$14,970	\$7,871	\$14,527
44	229	303	269	310	\$88,671	\$76,187	\$89,908	\$13,848	\$11,755	\$14,427

Officer 537 worked the most overtime in the data, and our estimates reveal this is due to both low access costs and a strong preference for overtime. As a result, officer 537 works similar amounts of overtime across all systems. Officer 537 is not very senior, and as a result is disadvantaged under a uniform-markdown auction and therefore prefers a uniform-wage auction. The effects of changing systems vary dramatically officer by officer, with some officers gaining surplus from moving away from informal trade and others losing. One officer works over 100 fewer shifts under a uniform-wage auction compared to informal trading, while another works almost 100 more. All members of the cartel would still earn more than \$50,000 under both counterfactual auctions, with several increasing their overtime pay.

Taken together, the above results suggest the cartel and the stark concentration of overtime it embodies would persist even if the overtime assignment system was centralized. This is because the large differences in overtime largely seem to reflect officer preference for work rather than differential access costs that are specific to the informal trade system.

F A Potential Bidding Ring

There is a “cartel” of highly connected officers who obtain many overtime shifts under the status quo informal trading system. Will this cartel form a bidding ring if the LADOT switched to an auction system? Several forces point to yes. Special events would need to be auctioned frequently, giving rise to repeat interaction. Fairness concerns would likely require the LADOT to use an auction format, like the uniform-wage or uniform markdown, which are known to be more susceptible to collusion than pay-your-wage bid alternatives (Robinson 1985). But several important forces point to no. First, there are many hundreds of bidders with highly dispersed valuations for overtime. Second, officers are geographically separated into divisions. Dispersion of values has been shown to make bidder collusion more difficult to maintain (Maskin and Riley 2000), while geographic separation makes communicating and enforcing complicated bidding strategies costly.²⁰

If the original cartel can capture a centralized auction, the wages and non-wage utility delivered discussed in Section 7 are upper bounds on the effects of auctions relative to the status quo. Still, there are ways to mitigate collusion while still addressing fairness concerns. One option would be to randomly draw officers to participate in each auction. If the set of participants changes over time, collusive strategies will be harder to maintain. Such a strategy of rotating who competes with whom was used successfully among chicken growers (Knoeber 1989).

²⁰The auctions we consider are multi-unit with single-unit demand. Marszalec, Teytelboym, and Laksá 2020 describe how many of the intuitions described in single-unit settings extend to multi-unit auctions with multi-unit demand, but to our knowledge there is less work examining multi-unit auctions with single-unit demand.